

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df

7730

Nombre de lignes df clean	7730
Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	5411
Taille dataset test	2319
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_12h_mean, r_NH3_st_12h_std, r_NH3_st_12h_rms, r_NH3_st_12h_median, r_NH3_st_12h_min, r_NH3_st_12h_max, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_temperature_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_rms, r_temperature_st_12h_median, r_temperature_st_12h_min, r_temperature_st_12h_max, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_umidity_st_12h_mean, r_umidity_st_12h_std, r_umidity_st_12h_rms, r_umidity_st_12h_median, r_umidity_st_12h_min, r_umidity_st_12h_max, r_umidity_st_12h_diff_last, r_umidity_st_12h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

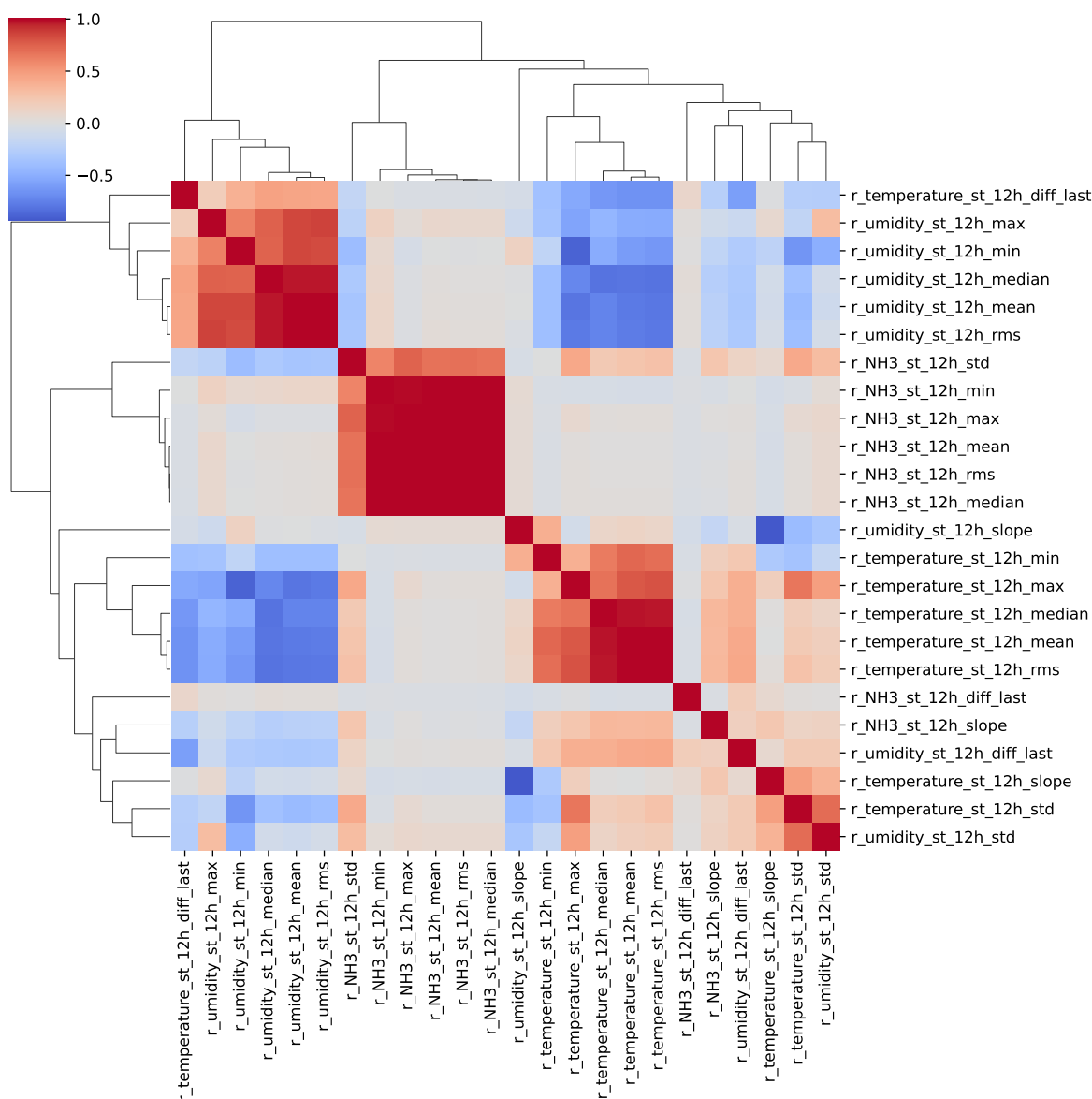
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_st_12h_median, r_NH3_st_12h_mean, r_NH3_st_12h_max, r_NH3_st_12h_rms, r_NH3_st_12h_min ; **cluster 7:** r_NH3_st_12h_std ; **cluster 8:** r_NH3_st_12h_diff_last

; **cluster 9:** r_NH3_st_12h_slope ; **cluster 10:** r_umidity_st_12h_mean, r_umidity_st_12h_min, r_umidity_st_12h_median, r_temperature_st_12h_max, r_temperature_st_12h_rms, r_temperature_st_12h_median, r_temperature_st_12h_mean, r_umidity_st_12h_max, r_umidity_st_12h_rms ; **cluster 11:** r_temperature_st_12h_std ; **cluster 12:** r_temperature_st_12h_min ; **cluster 13:** r_temperature_st_12h_diff_last ; **cluster 14:** r_temperature_st_12h_slope, r_umidity_st_12h_slope ; **cluster 15:** r_umidity_st_12h_std ; **cluster 16:** r_umidity_st_12h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Chooosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_12h_median, r_NH3_st_12h_std, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_umidity_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_min, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_umidity_st_12h_std, r_umidity_st_12h_diff_last



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_12h_median, r_NH3_st_12h_std, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_umidity_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_min, r_tem-

perature_st_12h_diff_last, r_temperature_st_12h_slope, r_umidity_st_12h_std,
r_umidity_st_12h_diff_last

Len(Features_X): 16

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model':
None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, ran-
dom_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

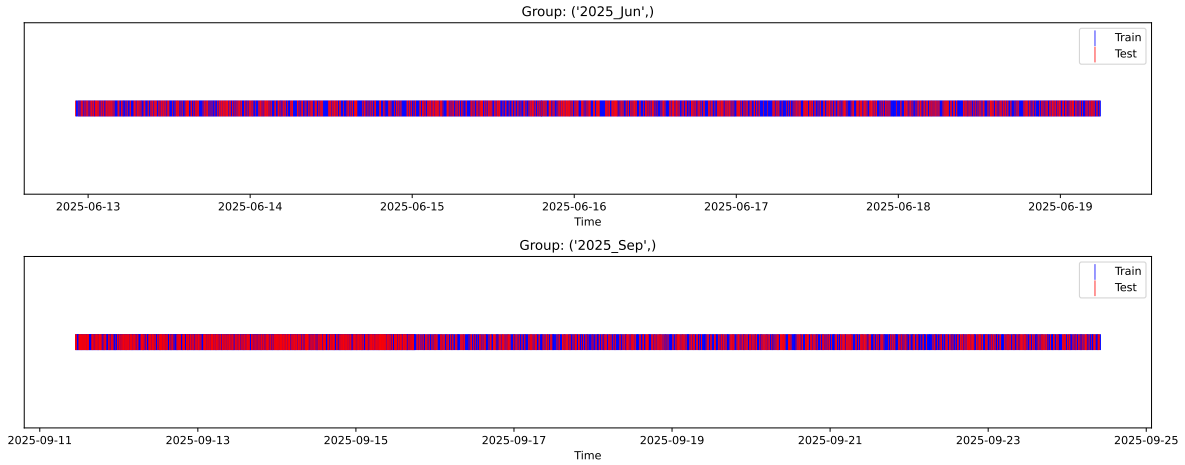
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5411 samples Test: 2319 samples



3.3.2 Internal CV Folds

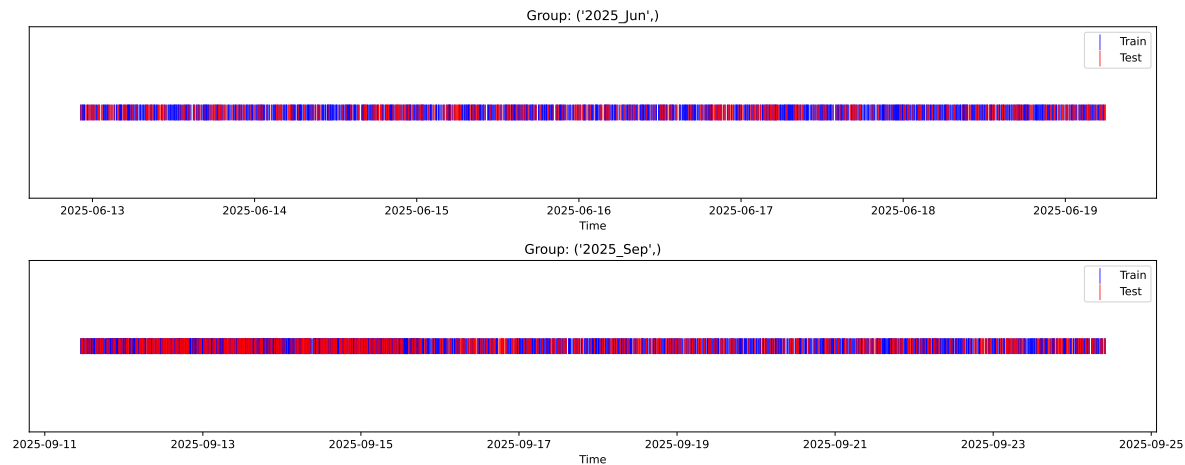
Fold 1/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



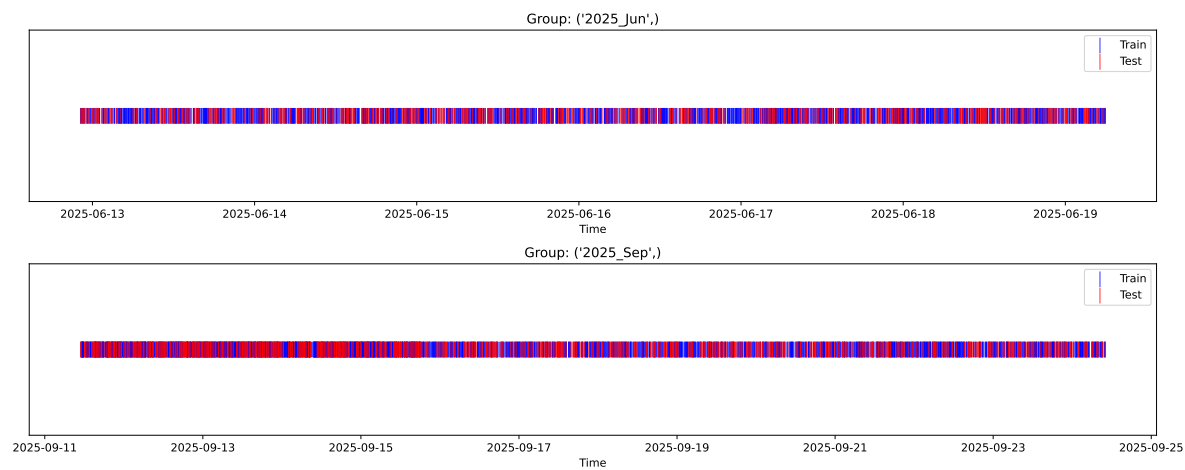
Fold 2/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



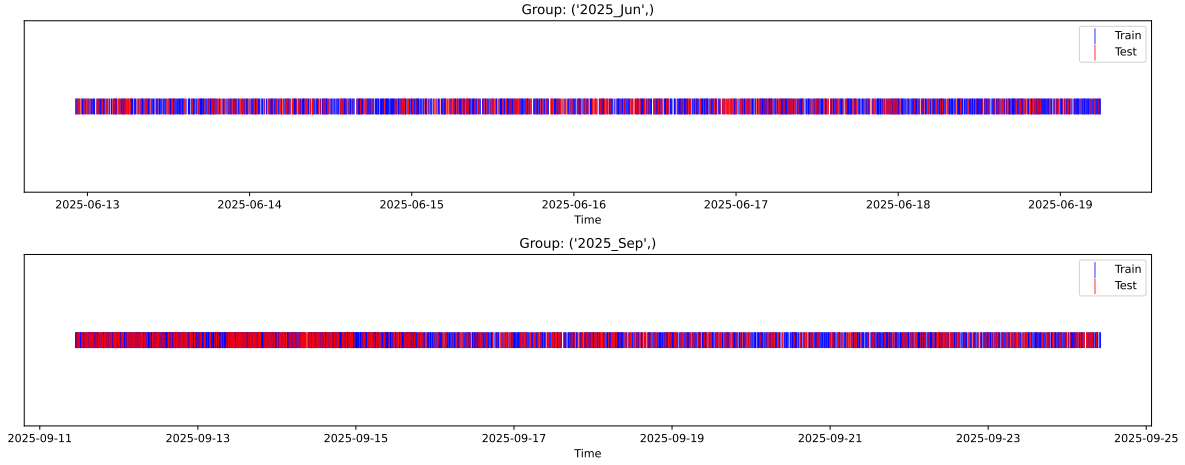
Fold 3/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



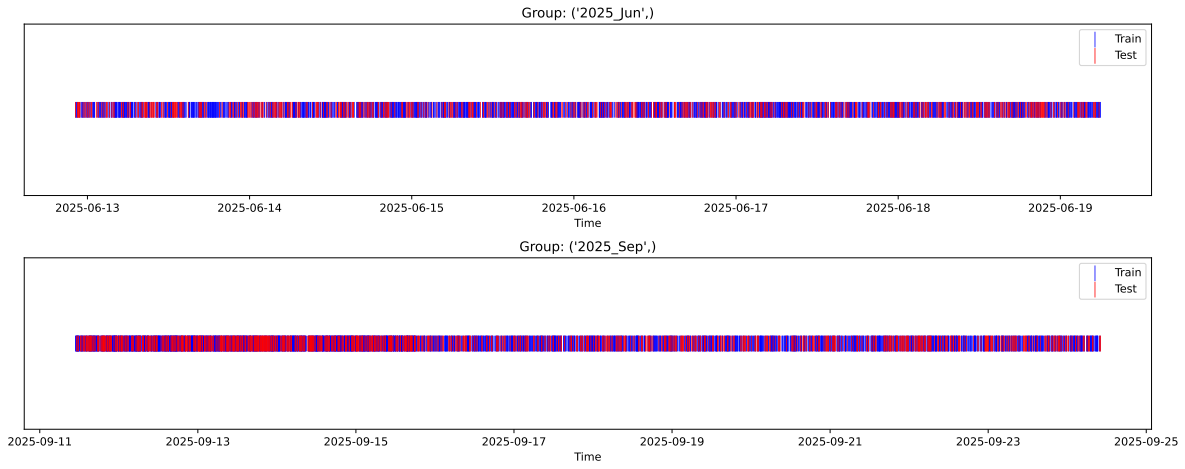
Fold 4/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.342, std=0.800, min=0.959, max=13.845
 y_{test} : mean=2.364, std=0.795, min=1.068, max=7.277

Fold 0 — y_{test} : mean=2.34, std=0.79, y_{train} : mean=2.34, std=0.80

Fold 1 — y_{test} : mean=2.35, std=0.83, y_{train} : mean=2.34, std=0.79

Fold 2 — y_{test} : mean=2.35, std=0.79, y_{train} : mean=2.34, std=0.81

Fold 3 — y_test: mean=2.33, std=0.77, y_train: mean=2.35, std=0.81

Fold 4 — y_test: mean=2.34, std=0.85, y_train: mean=2.34, std=0.78

train scores CV: [0.55549249 0.57679433 0.54833881 0.54990746 0.56328098]

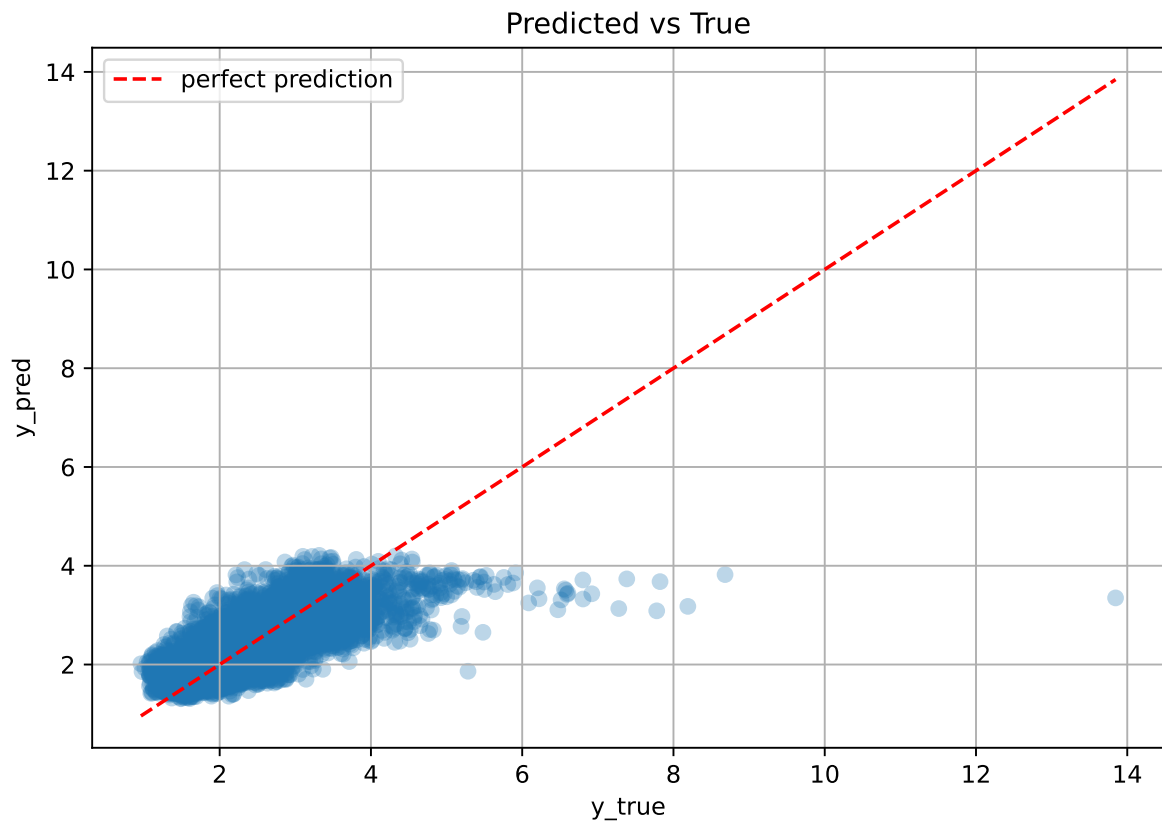
test scores CV: [0.54541905 0.50055427 0.56317109 0.55945045 0.53084447]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5588	0.5399	0.0227	0.5832	0.584	0.0441	1.0816	1.035

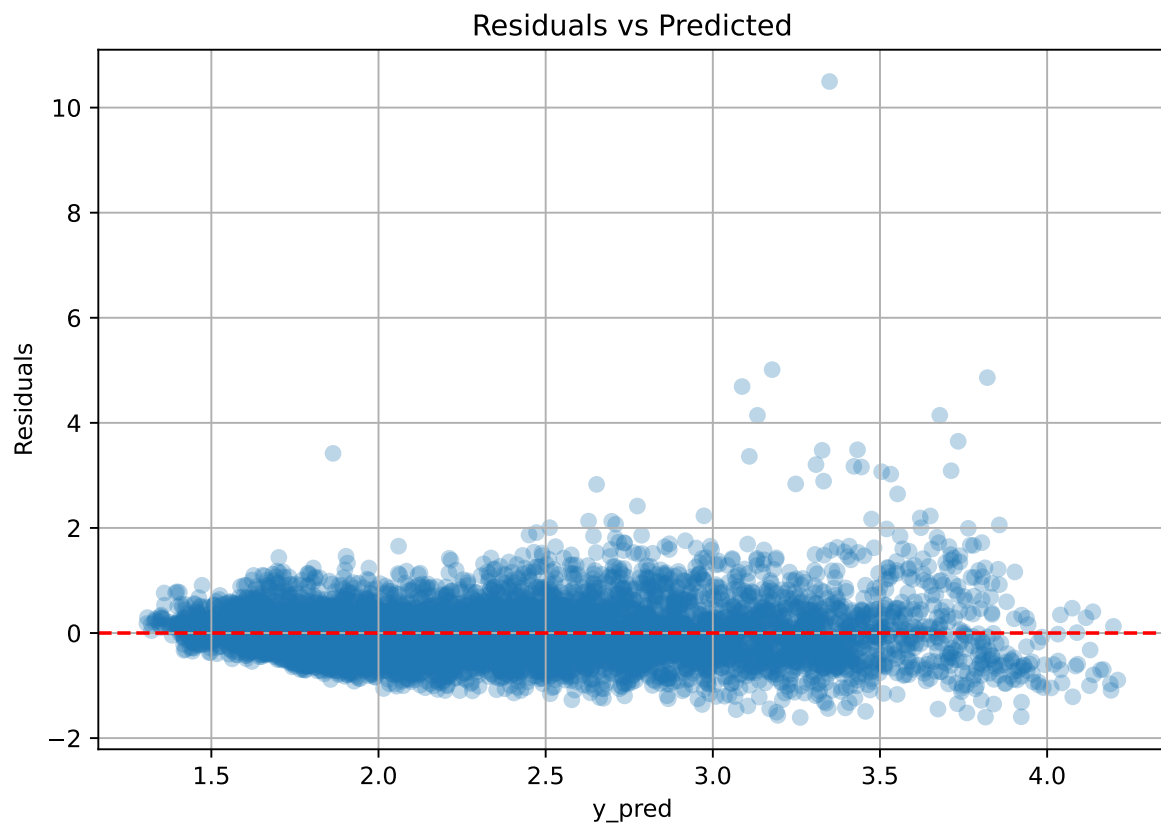
Coefficients:

coef_r_CO2 = -0.0460, coef_r_NH3 = 0.0331, coef_r_NH3_st_12h_diff_last = 0.0206, coef_r_NH3_st_12h_median = -0.0976, coef_r_NH3_st_12h_slope = 0.0344, coef_r_NH3_st_12h_std = 0.0322, coef_r_THI = 0.2853, coef_r_temperature = 0.4112, coef_r_temperature_st_12h_diff_last = 0.0007, coef_r_temperature_st_12h_min = -0.2499, coef_r_temperature_st_12h_slope = -0.6023, coef_r_temperature_st_12h_std = -0.1576, coef_r_umidity = -0.3913, coef_r_umidity_st_12h_diff_last = -0.0662, coef_r_umidity_st_12h_mean = 0.0435, coef_r_umidity_st_12h_std = -0.1232, intercept = 2.3417,

3.4.2 Scatter



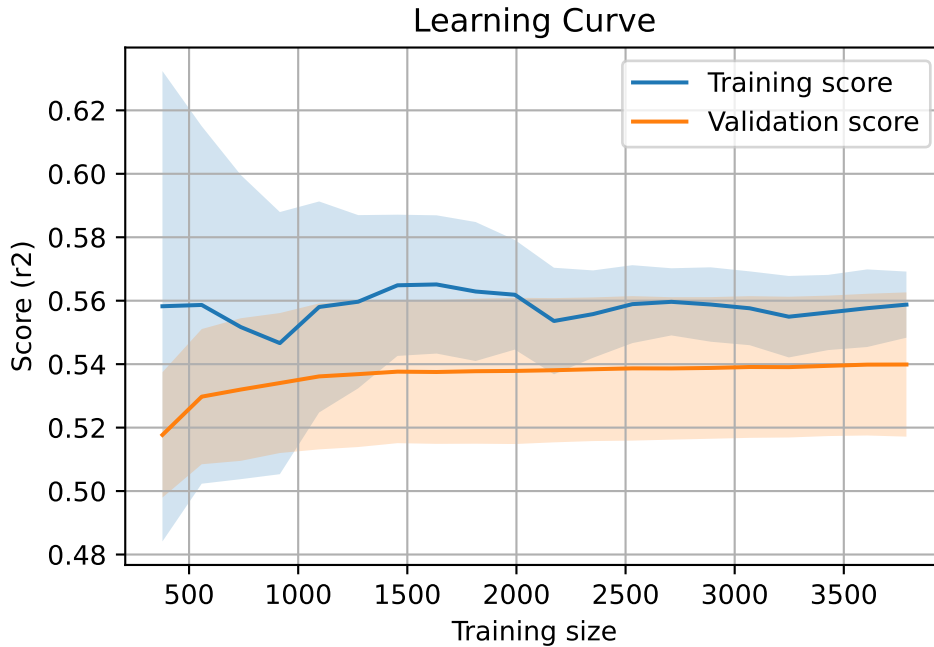
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7730, `Len(training)` : 5411,

`Len(training_real)` : 3788, `Len(test_of_training)` : 1623, `Len(test)` : 2319

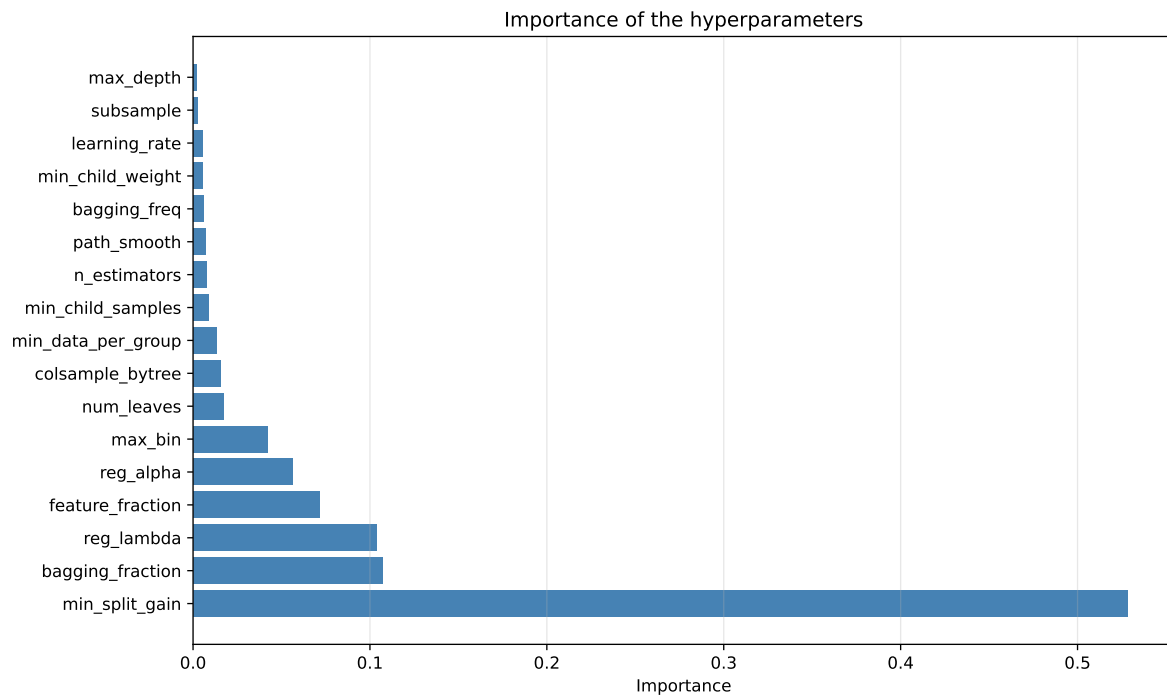
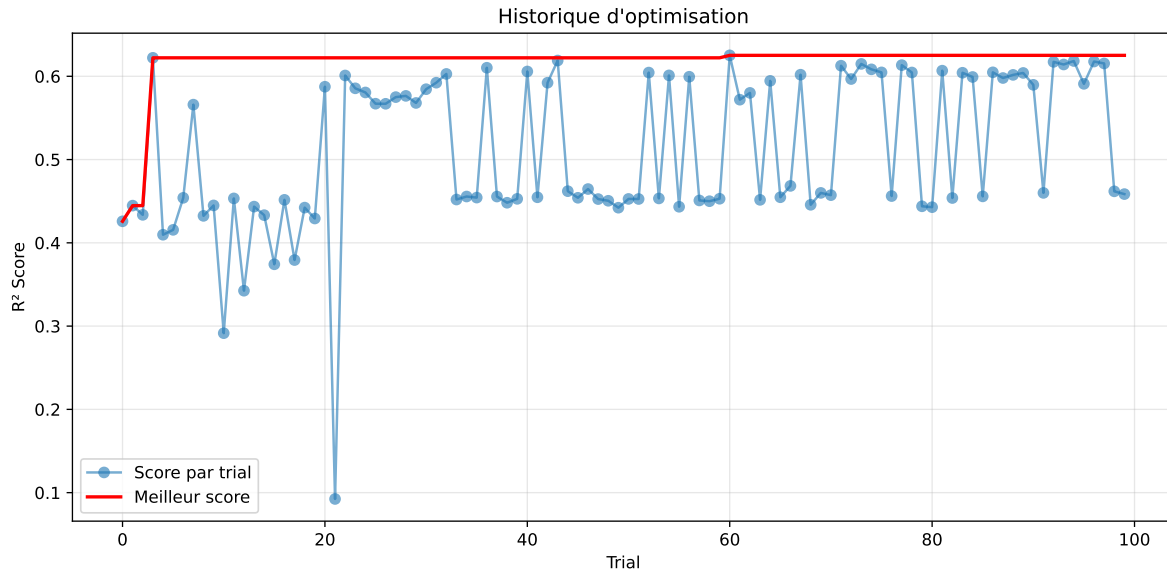


3.5 Model : LGBM

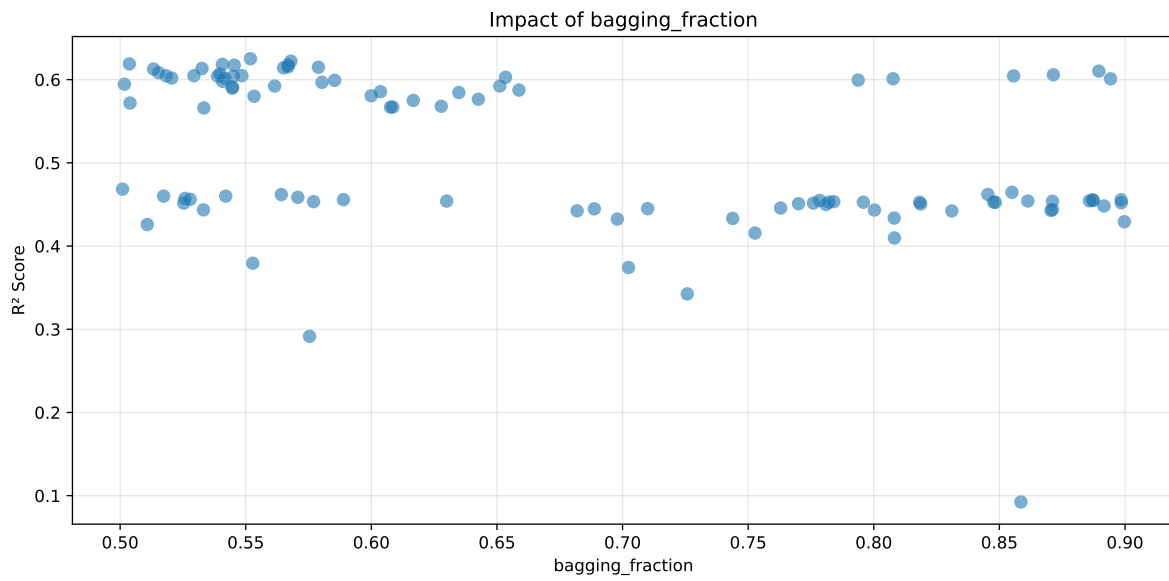
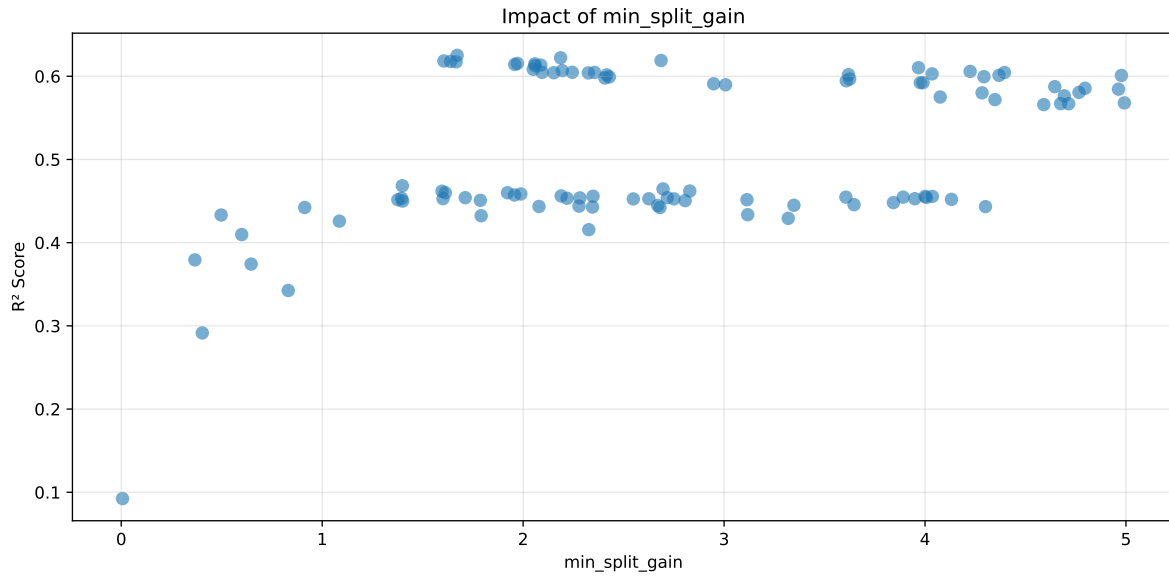
3.5.1 Gridsearch

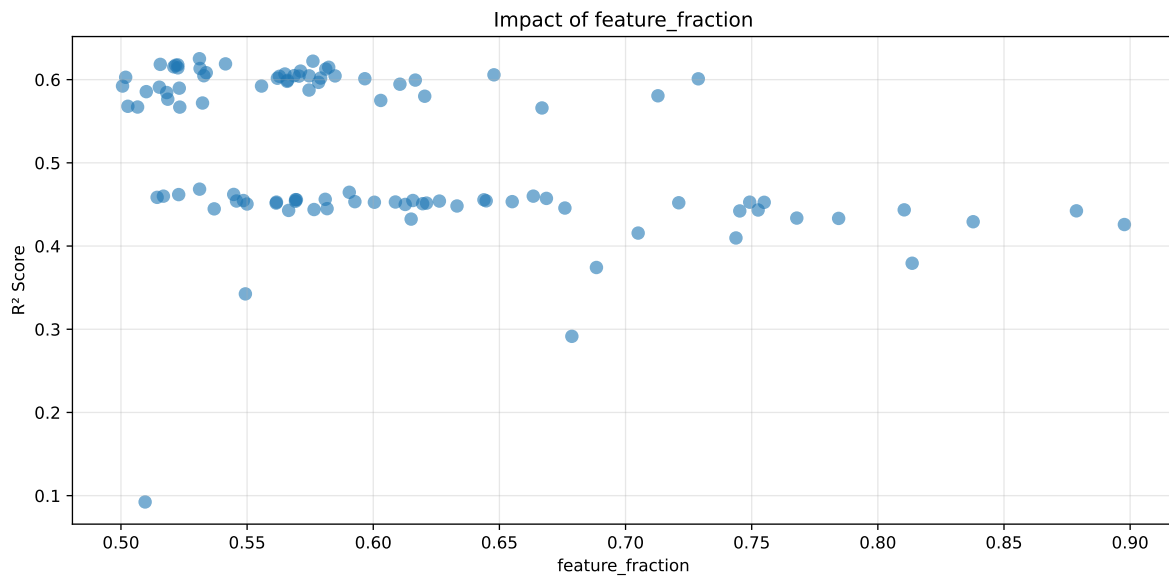
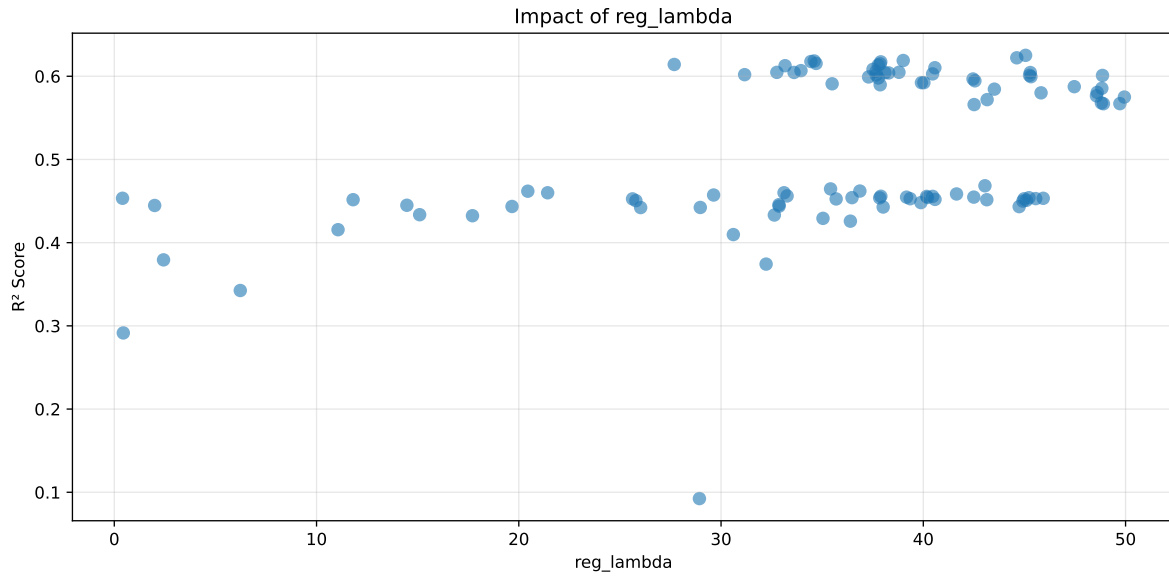
Distribution y_train vs y_test: y_train: mean=2.342, std=0.800, min=0.959, max=13.845
y_test: mean=2.364, std=0.795, min=1.068, max=7.277

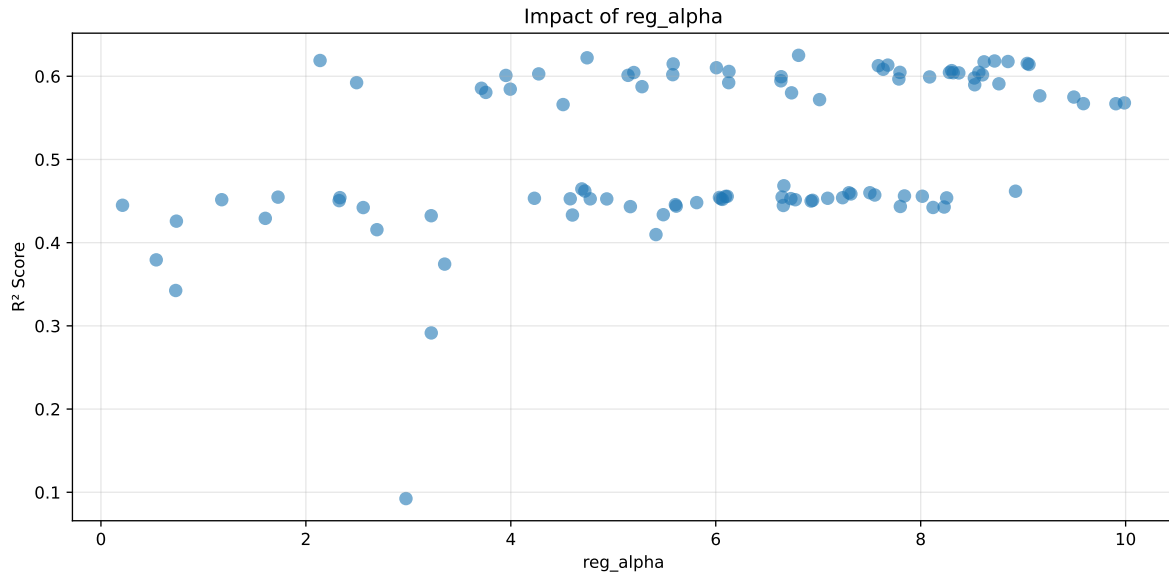
===== TOP Importance FEATURES =====
feature importance r_CO2 243 r_umidity 168 r_NH3_st_12h_median 153 r_tempera-
ture_st_12h_min 109 r_temperature_st_12h_diff_last 98 r_temperature 78 r_umid-
ity_st_12h_mean 75 r_umidity_st_12h_std 68 r_temperature_st_12h_slope 59 r_umid-
ity_st_12h_diff_last 59 r_NH3 57 r_NH3_st_12h_diff_last 43 r_temperature_st_12h_std
43 r_THI 41 r_NH3_st_12h_slope 37 r_NH3_st_12h_std 7



Paramètres avec >5% d'importance : ['min_split_gain', 'bagging_fraction', 'reg_lambda', 'feature_fraction', 'reg_alpha']

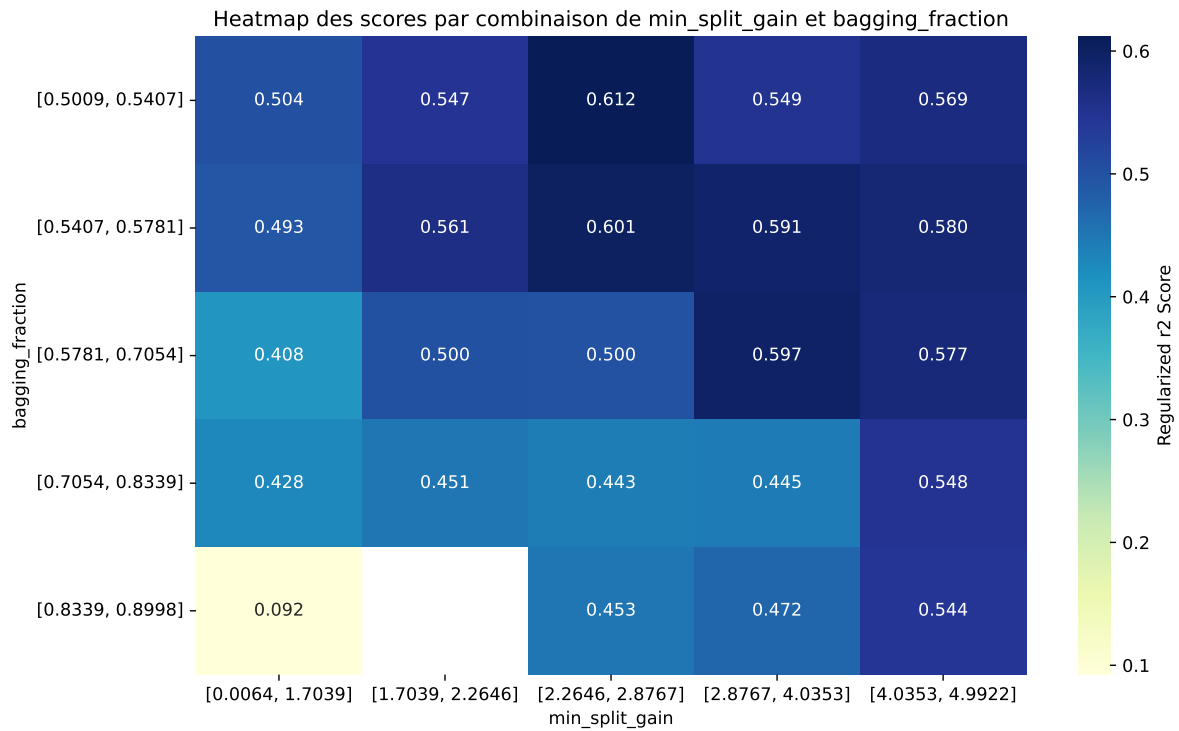






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.656	0.6251	0.6686	0.0273	0.6251	0.6986	0.0734	1.1175	1.0494
2	0.6507	0.6222	0.666	0.0291	0.6222	0.6872	0.065	1.1045	1.0458
3	0.6491	0.6189	0.6631	0.0271	0.6189	0.6877	0.0687	1.1111	1.0488
4	0.6484	0.6184	0.6612	0.0254	0.6184	0.6904	0.072	1.1165	1.0486
5	0.6484	0.6177	0.6603	0.0273	0.6177	0.6868	0.0691	1.1118	1.0497
6	0.648	0.6173	0.6615	0.0275	0.6173	0.687	0.0696	1.1128	1.0497
7	0.6459	0.6154	0.6587	0.0234	0.6154	0.6881	0.0727	1.1181	1.0495
8	0.6456	0.6149	0.6599	0.0269	0.6149	0.6767	0.0618	1.1005	1.0499
9	0.6447	0.6142	0.6577	0.0286	0.6142	0.6893	0.0752	1.1224	1.0498
10	0.6438	0.6134	0.6587	0.0274	0.6134	0.6771	0.0637	1.1039	1.0495
---	---	---	---	---	---	---	---	---	---
100	0.9078	0.7719	0.8189	0.0309	0.0924	0.837	0.065	1.0843	1.1761

Hyperparameters:

Rank 1: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.03333202491408125, 'reg__num_leaves': 59, 'reg__subsample': 0.8237972374495711, 'reg__colsample_bytree': 0.6025919203256767, 'reg__min_child_samples': 33, 'reg__reg_alpha': 6.8082756104545785, 'reg__reg_lambda': 45.05742481119076, 'reg__min_split_gain': 1.671578506417669, 'reg__path_smooth': 2.7259391326419617, 'reg__min_child_weight': 0.21554726327687534, 'reg__feature_fraction': 0.5311108681462633, 'reg__bagging_fraction': 0.5518582313003559, 'reg__bagging_freq': 2, 'reg__max_bin': 111, 'reg__min_data_per_group': 145}

Rank 2: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.05575744488503445, 'reg__num_leaves': 60, 'reg__subsample': 0.6975475208230055, 'reg__colsample_bytree': 0.5448475020658737, 'reg__min_child_samples': 33, 'reg__reg_alpha': 4.744286124747159, 'reg__reg_lambda': 44.621727534702785, 'reg__min_split_gain': 2.1864532914151025, 'reg__path_smooth': 4.217042948247222, 'reg__min_child_weight': 0.5458229345426545, 'reg__feature_fraction': 0.5760839080916877, 'reg__bagging_fraction': 0.5679490179794027, 'reg__bagging_freq': 3, 'reg__max_bin': 68, 'reg__min_data_per_group': 97}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.0335477544501681, 'reg__num_leaves': 19, 'reg__subsample': 0.6803409555094759, 'reg__colsample_bytree': 0.6104394691340583, 'reg__min_child_samples': 26, 'reg__reg_alpha': 2.138408910857079, 'reg__reg_lambda': 39.007060313515964, 'reg__min_split_gain': 2.6864447518921164, 'reg__path_smooth': 0.10812396551159464, 'reg__min_child_weight': 0.0036642305533044995, 'reg__feature_fraction': 0.5414413275694582, 'reg__bagging_fraction': 0.5036611530235153, 'reg__bagging_freq': 3, 'reg__max_bin': 84, 'reg__min_data_per_group': 104}

Rank 4: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.043272979041545846, 'reg__num_leaves': 46, 'reg__subsample': 0.8969565565301248, 'reg__colsample_bytree': 0.5884330864688591, 'reg__min_child_samples': 41, 'reg__reg_alpha': 8.721016481082916, 'reg__reg_lambda': 34.6081052358344, 'reg__min_split_gain': 1.605551802245289, 'reg__path_smooth': 3.8243515218556046, 'reg__min_child_weight': 0.5839295197006598, 'reg__feature_fraction': 0.5155906666758721, 'reg__bagging_fraction': 0.5406861100658209, 'reg__bagging_freq': 4, 'reg__max_bin': 147, 'reg__min_data_per_group': 138}

Rank 5: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.060214944349084615, 'reg__num_leaves': 39, 'reg__subsample': 0.7550274657992422, 'reg__colsample_bytree': 0.559613427521457, 'reg__min_child_samples': 41, 'reg__reg_alpha': 8.852115866512067, 'reg__reg_lambda': 34.44241932056158, 'reg__min_split_gain': 1.6389866603237813, 'reg__path_smooth': 3.8035510977054754, 'reg__min_child_weight': 0.5711909150561775, 'reg__feature_fraction': 0.5225474773317342, 'reg__bagging_fraction': 0.566945748162906, 'reg__bagging_freq': 4, 'reg__max_bin': 148, 'reg__min_data_per_group': 137}

Rank 6: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.042612866197007646, 'reg__num_leaves': 40, 'reg__subsample': 0.8765331175464127, 'reg__colsample_bytree': 0.5116800908774181, 'reg__min_child_samples': 39, 'reg__reg_alpha': 8.618315835722868, 'reg__reg_lambda': 37.89674627156569, 'reg__min_split_gain': 1.666276076593229, 'reg__path_smooth': 3.8223238217988564, 'reg__min_child_weight': 1.0832005331035623, 'reg__feature_fraction': 0.521673898939526, 'reg__bagging_fraction': 0.5454078370785842, 'reg__bagging_freq': 4, 'reg__max_bin': 144, 'reg__min_data_per_group': 163}

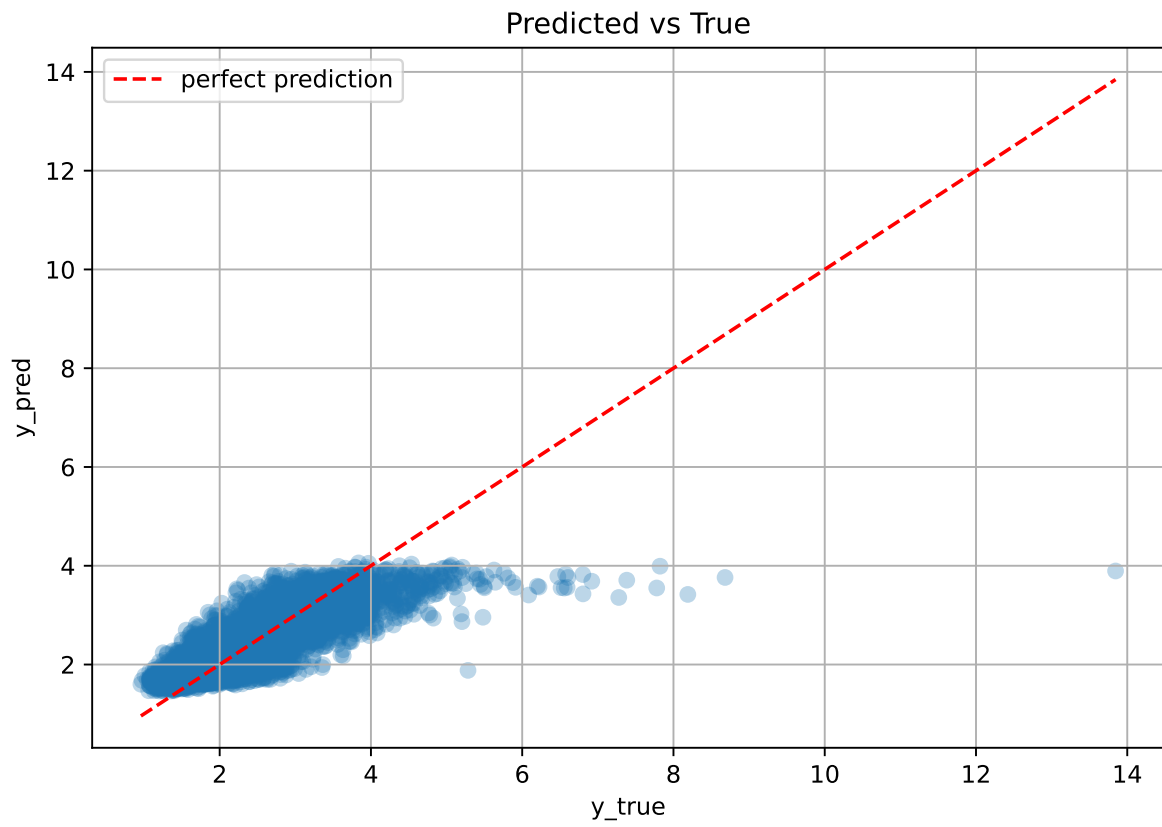
Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.04271490717183721, 'reg__num_leaves': 46, 'reg__subsample': 0.7279619852052799, 'reg__colsample_bytree': 0.5588621706127651, 'reg__min_child_samples': 41, 'reg__reg_alpha': 9.040002860204394, 'reg__reg_lambda': 34.69838805904621, 'reg__min_split_gain': 1.9718722285820258, 'reg__path_smooth': 4.277899895861626, 'reg__min_child_weight': 1.1910464362091766, 'reg__feature_fraction': 0.5209513704902408, 'reg__bagging_fraction': 0.5668369636592038, 'reg__bagging_freq': 3, 'reg__max_bin': 154, 'reg__min_data_per_group': 138}

Rank 8: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.05731378519759158, 'reg__num_leaves': 41, 'reg__subsample': 0.7167423328336131, 'reg__colsample_bytree': 0.5731427030627083, 'reg__min_child_samples': 43, 'reg__reg_alpha': 5.584482578124877, 'reg__reg_lambda': 37.865747461692976, 'reg__min_split_gain': 2.0580177558645767, 'reg__path_smooth': 1.6691918595458701, 'reg__min_child_weight': 0.7195796793520233, 'reg__feature_fraction': 0.5823455560617687, 'reg__bagging_fraction': 0.5789168968574923, 'reg__bagging_freq': 3, 'reg__max_bin': 120, 'reg__min_data_per_group': 144}

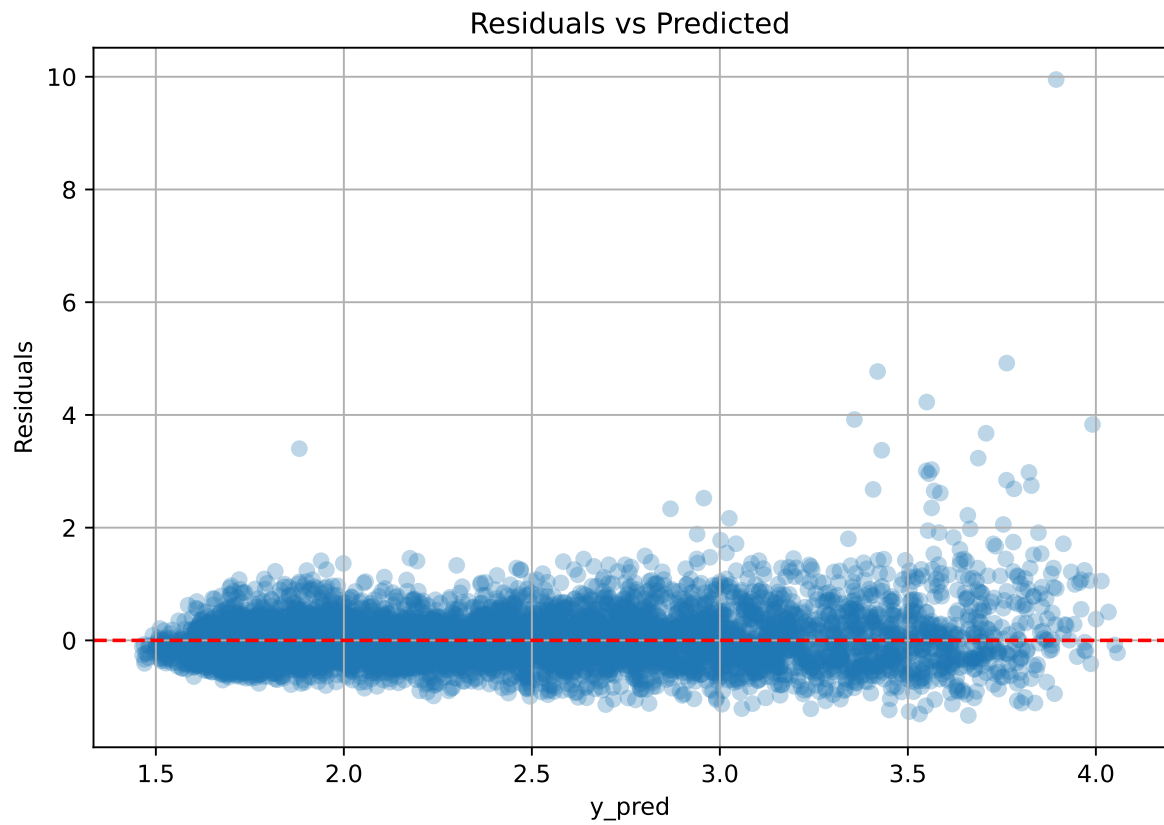
Rank 9: {'reg__n_estimators': 200, 'reg__max_depth': 8, 'reg__learning_rate': 0.04146237846043251, 'reg__num_leaves': 39, 'reg__subsample': 0.8838564617332776, 'reg__colsample_bytree': 0.5308075434269837, 'reg__min_child_samples': 31, 'reg__reg_alpha': 9.057137563477507, 'reg__reg_lambda': 27.685046597717545, 'reg__min_split_gain': 1.9582566762558544, 'reg__path_smooth': 4.402207632239463, 'reg__min_child_weight': 0.5561378703084433, 'reg__feature_fraction': 0.5225545295873603, 'reg__bagging_fraction': 0.5650457597577259, 'reg__bagging_freq': 4, 'reg__max_bin': 145, 'reg__min_data_per_group': 136}

Rank 10: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.05562083136032541, 'reg__num_leaves': 42, 'reg__subsample': 0.7148871984634475, 'reg__colsample_bytree': 0.5754495876258517, 'reg__min_child_samples': 39, 'reg__reg_alpha': 7.679193170886444, 'reg__reg_lambda': 37.774228317302004, 'reg__min_split_gain': 2.087803050597687, 'reg__path_smooth': 3.4335359643879824, 'reg__min_child_weight': 0.7647333763381114, 'reg__feature_fraction': 0.5313602575474894, 'reg__bagging_fraction': 0.5325755035436075, 'reg__bagging_freq': 1, 'reg__max_bin': 125, 'reg__min_data_per_group': 146}

3.5.2 Scatter



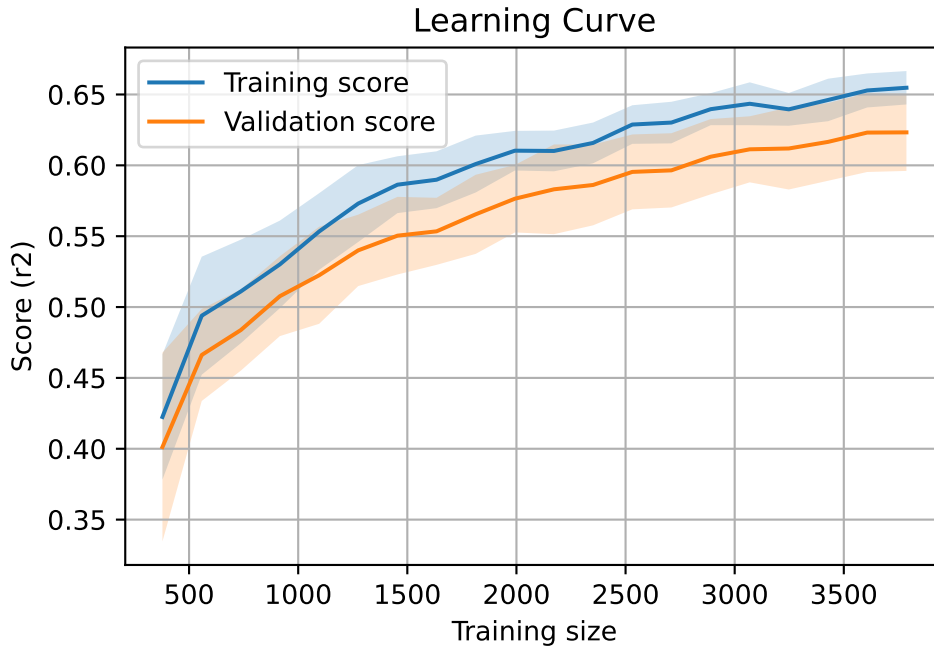
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7730, `Len(training)` : 5411,

`Len(training_real)` : 3788, `Len(test_of_training)` : 1623, `Len(test)` : 2319



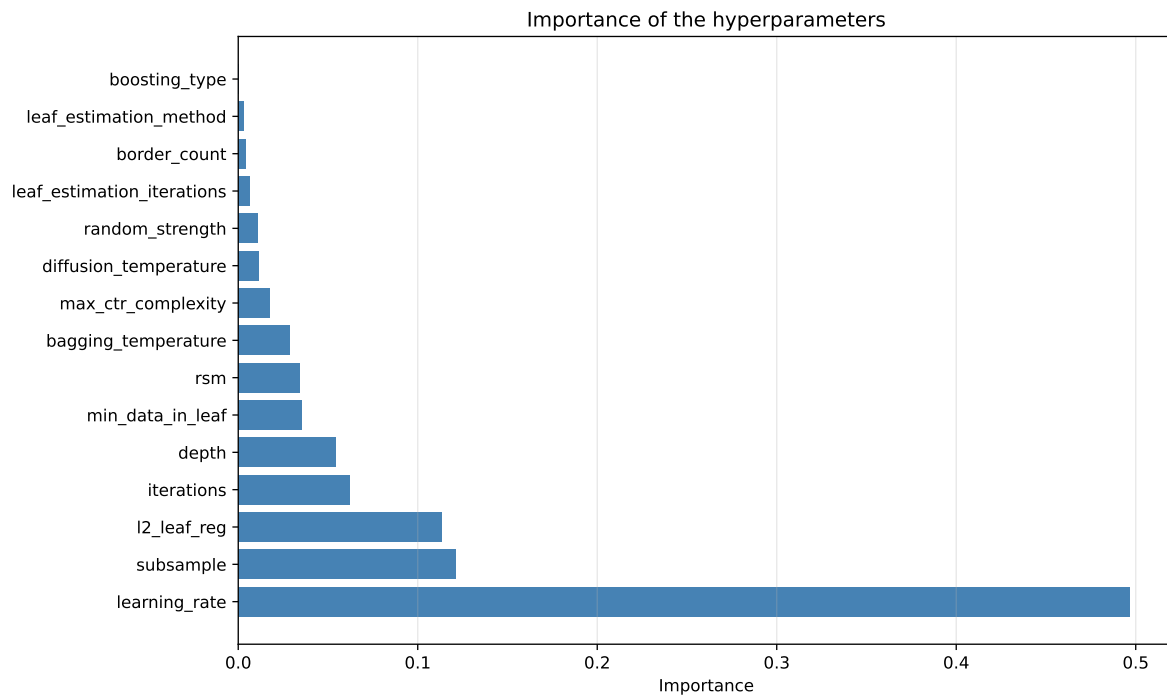
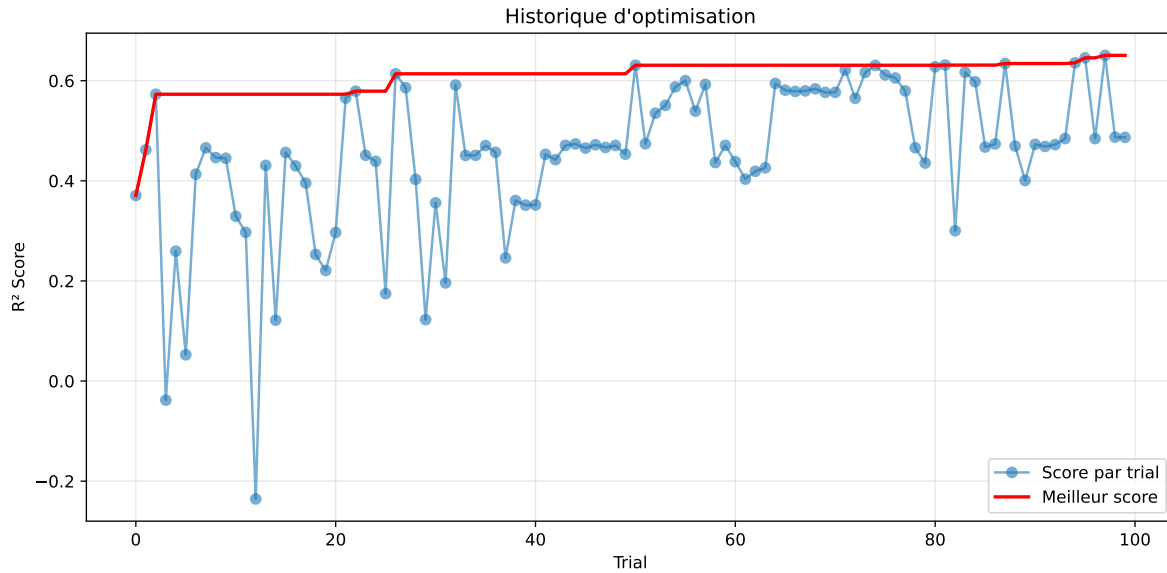
3.6 Model : CatBoost

3.6.1 Gridsearch

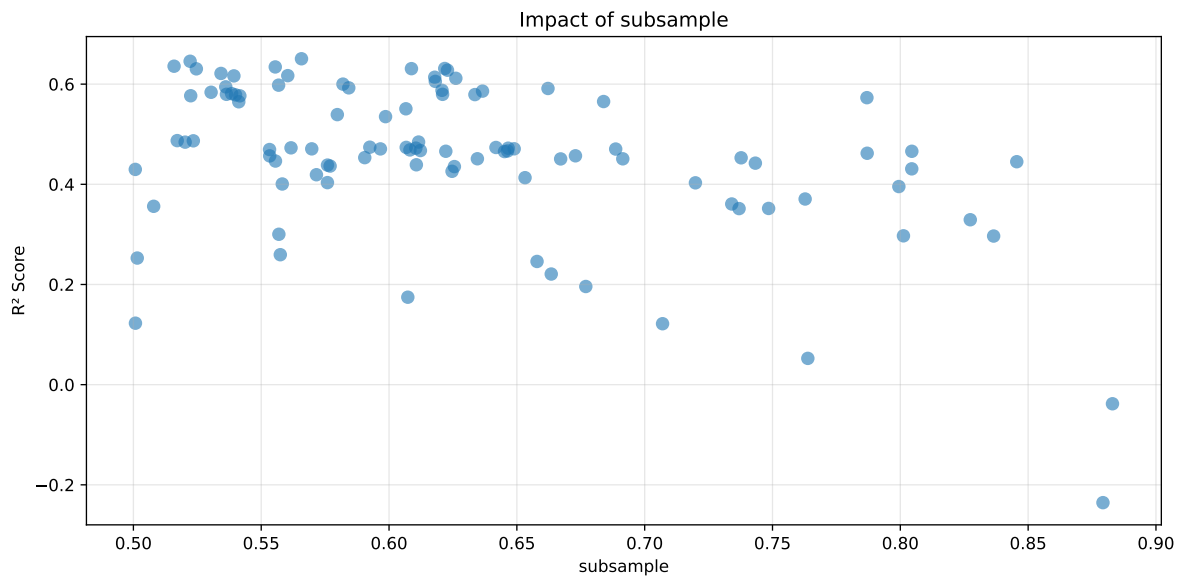
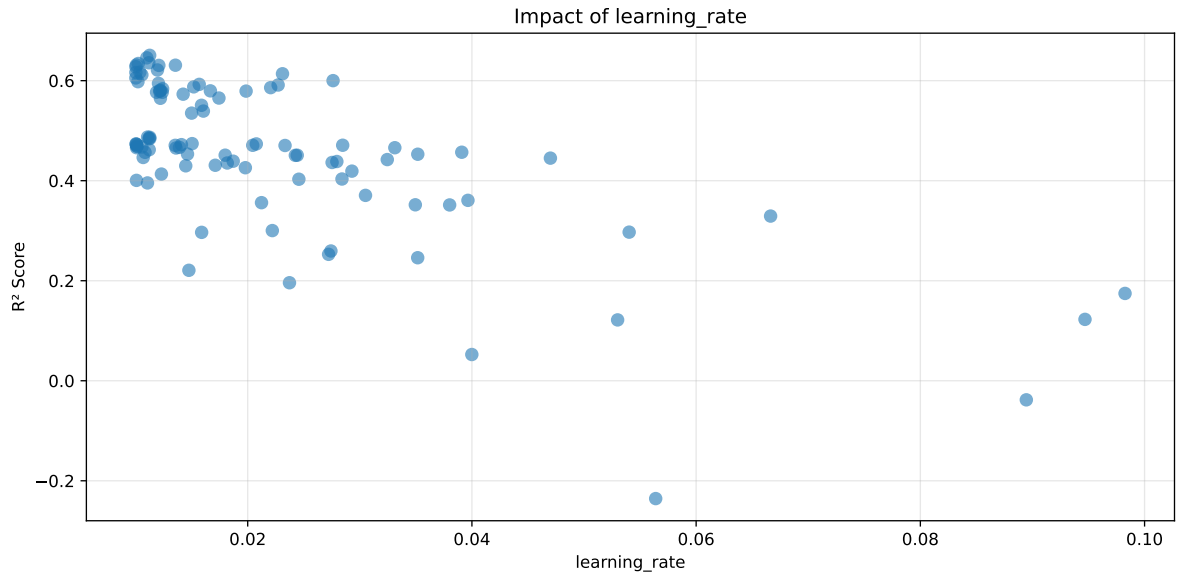
Distribution y_train vs y_test: y_train: mean=2.342, std=0.800, min=0.959, max=13.845
y_test: mean=2.364, std=0.795, min=1.068, max=7.277

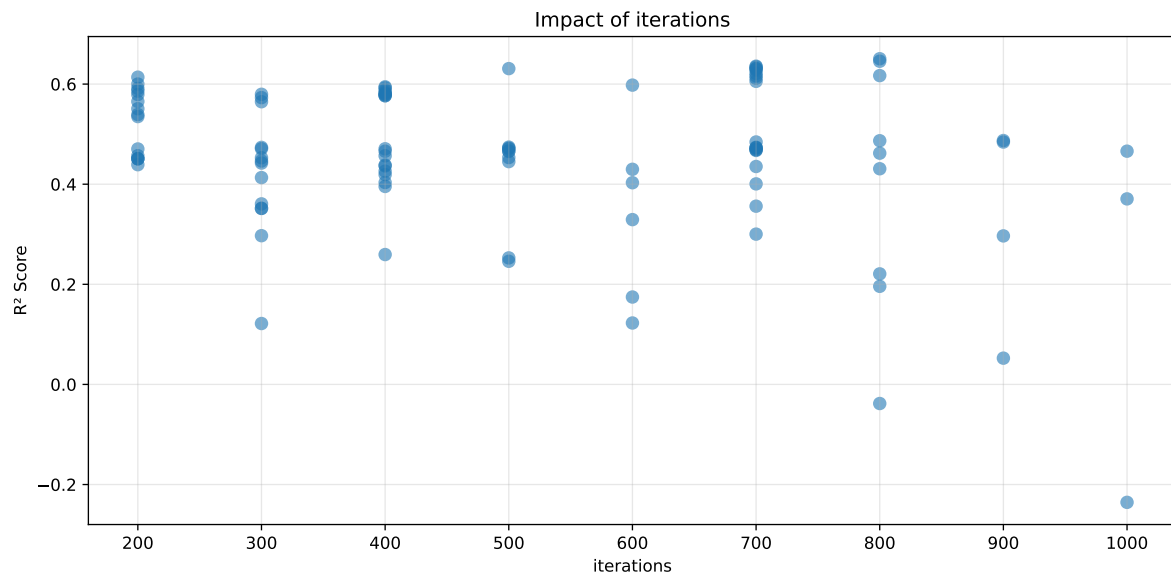
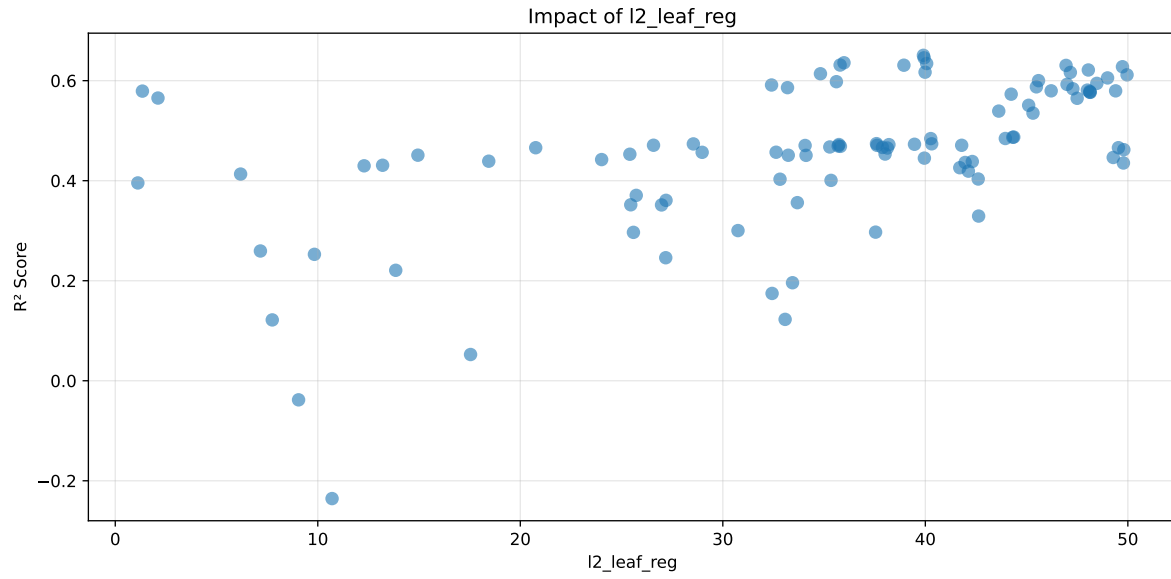
===== TOP Importance FEATURES =====

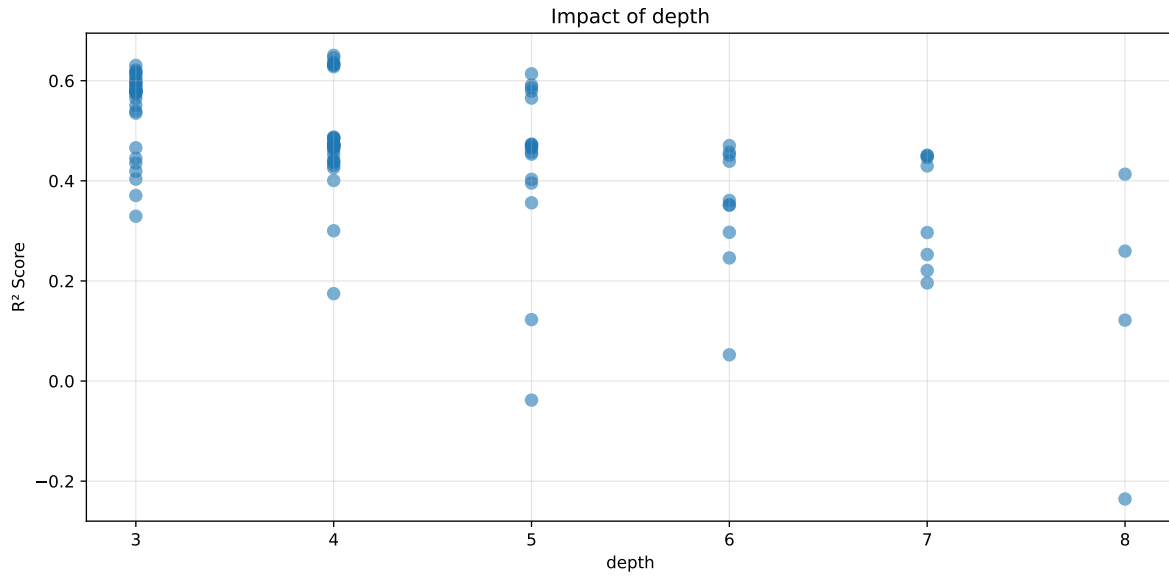
Feature	Importance
r_umidity	27.461812
r_temperature_st_12h_diff_last	16.320561
r_temperature	8.834341
r_CO2	7.435523
r_NH3_st_12h_median	6.307986
r_umidity_st_12h_std	4.967701
r_umidity_st_12h_diff_last	4.524632
r_THI	4.454049
r_umidity_st_12h_mean	3.770680
r_temperature_st_12h_slope	3.174374
r_temperature_st_12h_min	3.059832
r_NH3	2.930139
r_NH3_st_12h_slope	2.493623
r_temperature_st_12h_std	1.943160
r_NH3_st_12h_std	1.431974
r_NH3_st_12h_diff_last	0.889614



Paramètres avec >5% d'importance : ['learning_rate', 'subsample', 'l2_leaf_reg', 'iterations', 'depth']

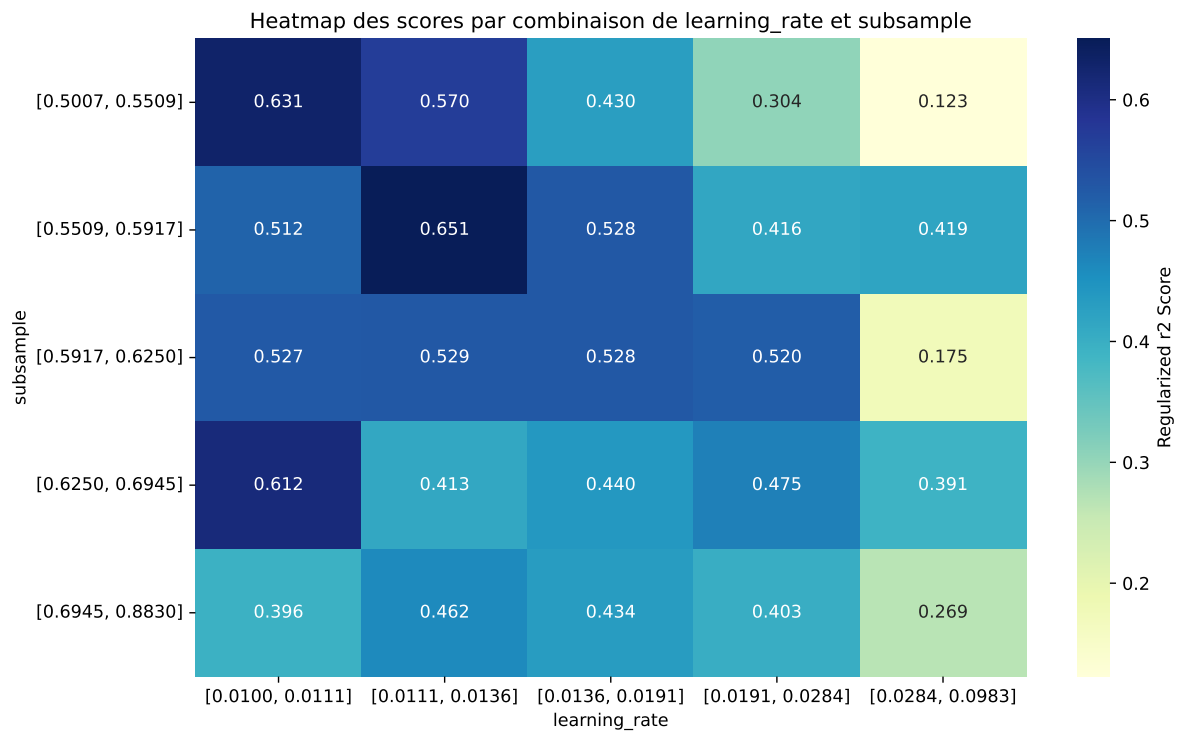






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6826	0.6507	0.6974	0.0247	0.6507	0.7034	0.0528	1.0811	1.049
2	0.677	0.6457	0.6923	0.0245	0.6457	0.7013	0.0557	1.0862	1.0485
3	0.6648	0.6358	0.6818	0.0241	0.6358	0.6891	0.0533	1.0838	1.0457
4	0.6653	0.6343	0.6806	0.0236	0.6343	0.6881	0.0538	1.0848	1.0488
5	0.6624	0.6312	0.6776	0.0235	0.6312	0.6847	0.0534	1.0847	1.0494
6	0.6607	0.6309	0.6761	0.0244	0.6309	0.6827	0.0517	1.082	1.0472
7	0.6584	0.6305	0.6763	0.0251	0.6305	0.683	0.0525	1.0833	1.0441
8	0.658	0.6279	0.6736	0.0241	0.6279	0.68	0.0521	1.083	1.048
9	0.6466	0.6214	0.6673	0.0242	0.6214	0.6763	0.0549	1.0883	1.0406
10	0.6442	0.6169	0.6634	0.0241	0.6169	0.6695	0.0526	1.0852	1.0442
---	---	---	---	---	---	---	---	---	---
100	0.994	0.7891	0.8346	0.0322	-0.2356	0.856	0.0669	1.0848	1.2597

Hyperparameters:

Rank 1: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.011255944377604406, 'reg__l2_leaf_reg': 39.905378263184076, 'reg__bagging_temperature': 3.045023804714705, 'reg__random_strength': 3.4286868817304526, 'reg__subsample': 0.5657515779650027, 'reg__min_data_in_leaf': 47, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.5916361051827114, 'reg__border_count': 127, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 5240.484910976321}

Rank 2: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.010999548978492346, 'reg__l2_leaf_reg': 39.94046358034814, 'reg__bagging_temperature': 3.5132550814862684, 'reg__random_strength': 3.4730928669960592, 'reg__subsample': 0.5221908338292839, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.5762141941648502, 'reg__border_count': 125, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 5296.113307024605}

Rank 3: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.011182138625352715, 'reg__l2_leaf_reg': 35.983591374948396, 'reg__bagging_temperature': 3.021027830830115, 'reg__random_strength': 3.8231898494615075, 'reg__subsample': 0.5159124967281882, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5979750152820875, 'reg__border_count': 125, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9928.642733884111}

Rank 4: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.01024145139235195, 'reg__l2_leaf_reg': 40.06504093970277, 'reg__bagging_temperature': 2.6958151541797917, 'reg__random_strength': 3.321037247605174, 'reg__subsample': 0.5554921001991352, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 9, 'reg__rsm':

0.5946669708440161, 'reg__border_count': 124, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 5612.205383838614}

Rank 5: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.010065791936534434, 'reg__l2_leaf_reg': 35.791891879126624, 'reg__bagging_temperature': 3.5424990569175407, 'reg__random_strength': 3.402018848934617, 'reg__subsample': 0.6217842570415606, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5414922130571852, 'reg__border_count': 103, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5577.273240505569}

Rank 6: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.013553138407018404, 'reg__l2_leaf_reg': 38.941093550596804, 'reg__bagging_temperature': 2.303891909802663, 'reg__random_strength': 3.166423078733644, 'reg__subsample': 0.6087658035880381, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.5225480293748372, 'reg__border_count': 64, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 6747.177875631544}

Rank 7: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.012073767600270763, 'reg__l2_leaf_reg': 46.9483019204577, 'reg__bagging_temperature': 1.3646062455156849, 'reg__random_strength': 1.701815175214257, 'reg__subsample': 0.5246233567514459, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5179709123404718, 'reg__border_count': 124, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 6142.495681354535}

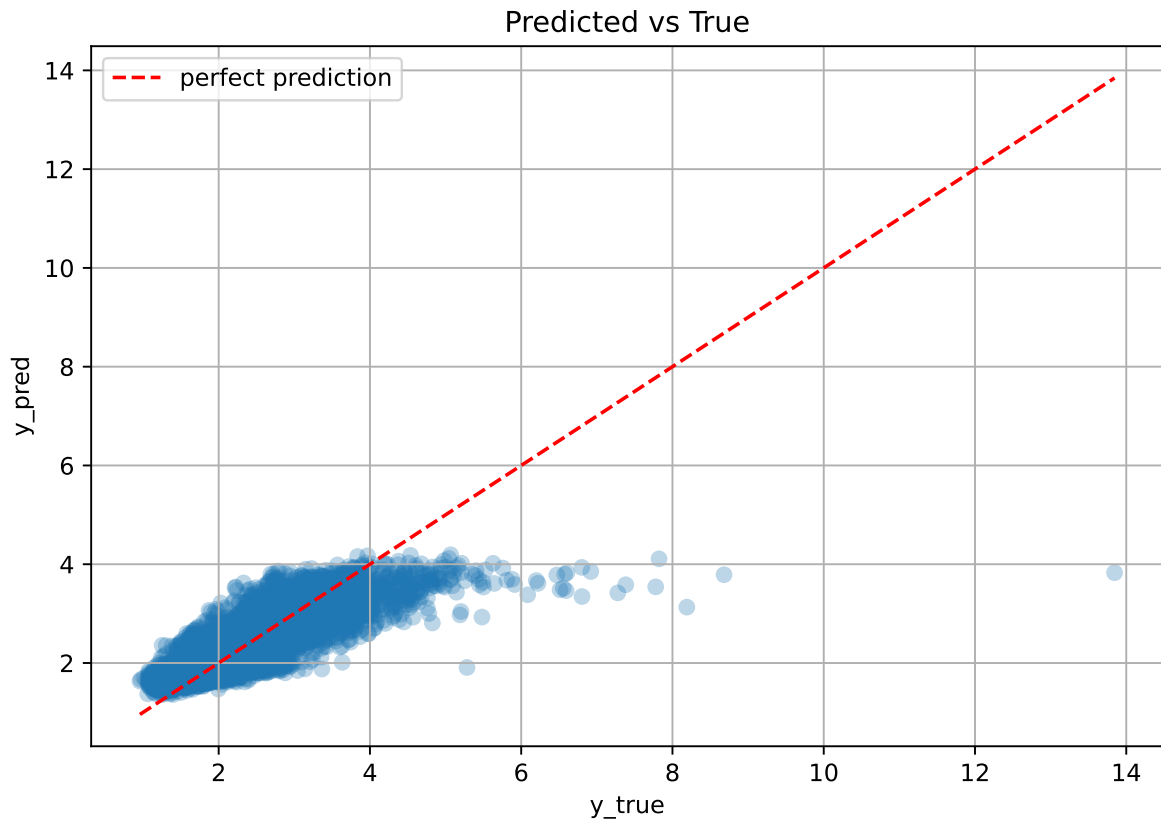
Rank 8: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.010023044625661517, 'reg__l2_leaf_reg': 49.734207570686685, 'reg__bagging_temperature': 2.6955341567559845, 'reg__random_strength': 3.4577376657003773, 'reg__subsample': 0.6228558212223025, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5459706029742909, 'reg__border_count': 45, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5494.525436731283}

Rank 9: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.011950606262311467, 'reg__l2_leaf_reg': 48.048792199119276, 'reg__bagging_temperature': 1.383259561297049, 'reg__random_strength': 2.430515662445167, 'reg__subsample': 0.5342655787695257, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5164443236917674, 'reg__border_count': 110, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9199.667344228277}

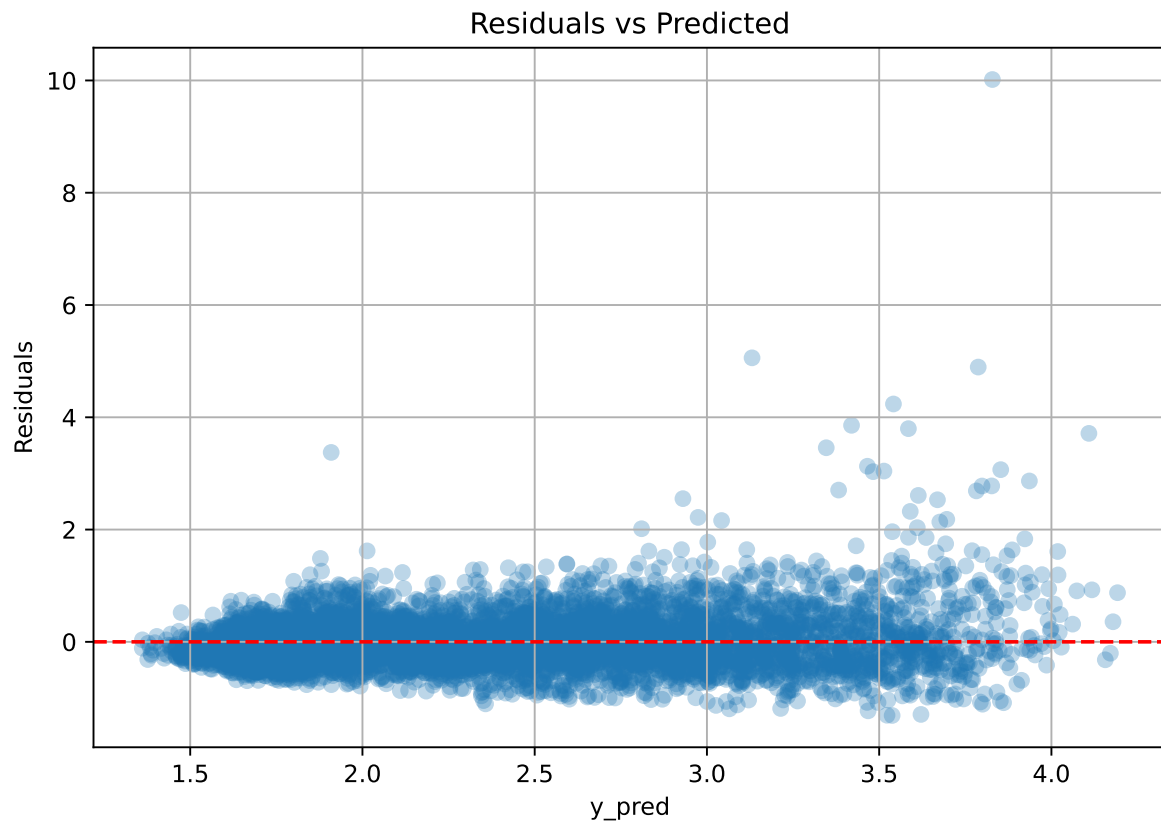
Rank 10: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.01038208889495651, 'reg__l2_leaf_reg': 39.987008308398075, 'reg__bagging_temperature': 2.684649960874445,

'reg__random_strength': 3.390738835612954, 'reg__subsample': 0.560387728974413,
'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 9, 'reg__rsm':
0.5439565934387547, 'reg__border_count': 124, 'reg__leaf_estimation_method': 'Gradient',
'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature':
9690.016809035846}

3.6.2 Scatter



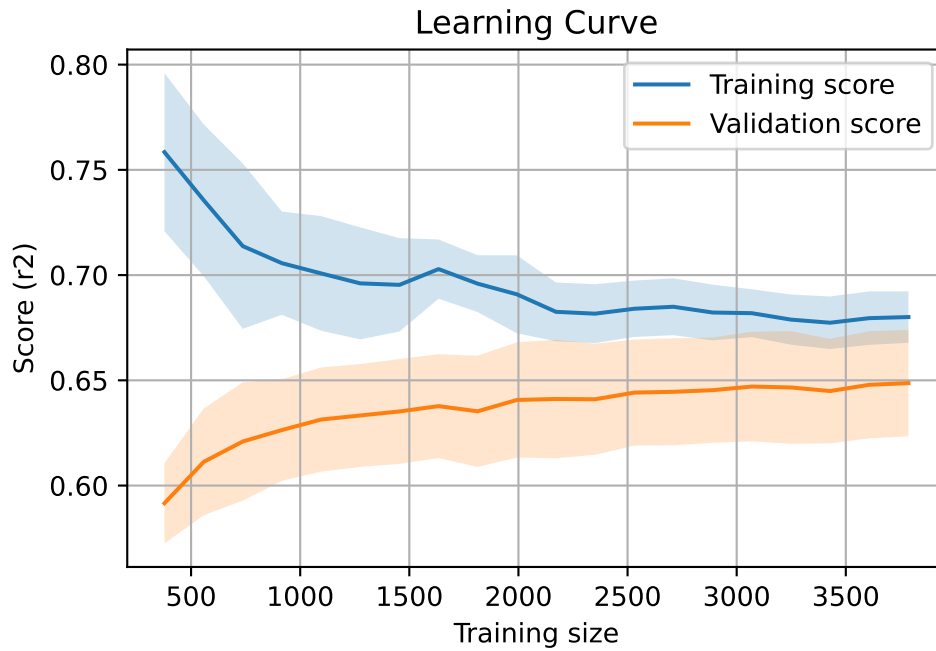
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7730, `Len(training)` : 5411,

`Len(training_real)` : 3788, `Len(test_of_training)` : 1623, `Len(test)` : 2319



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.559	0.54	0.023	0.583	0.584	0.044	1.082	1.035	nan
LGBM	0.656	0.625	0.027	0.669	0.699	0.073	1.118	1.049	0.6251
CatBoost	0.683	0.651	0.025	0.697	0.703	0.053	1.081	1.049	0.6507

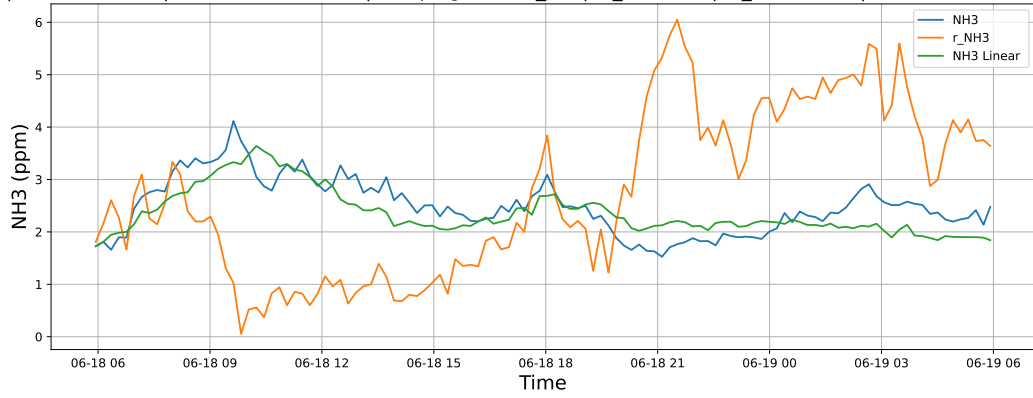
4 Plot

4.1 Standalone plots

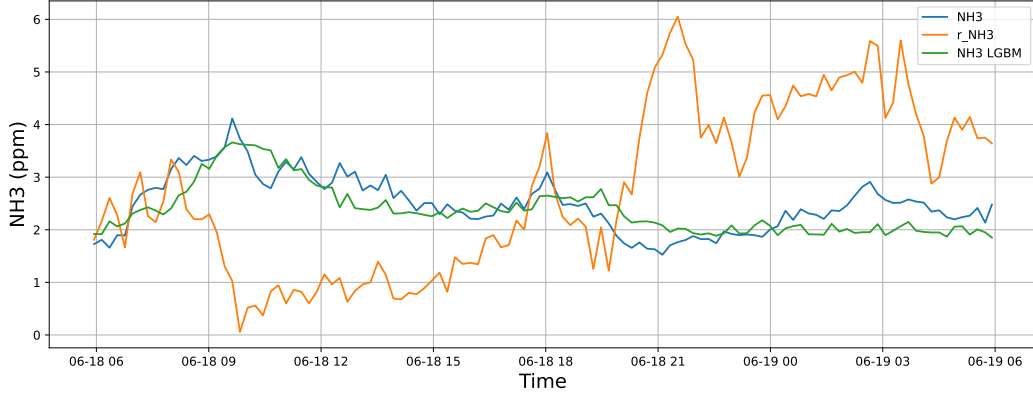
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

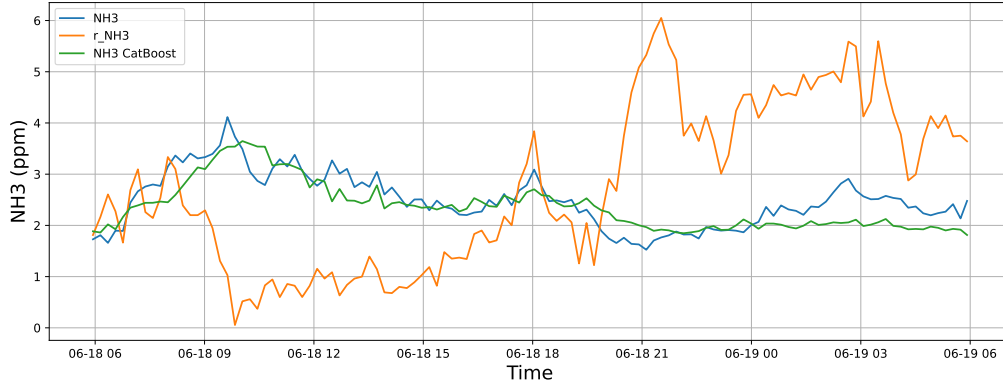
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



4.1.1.2 Time window : 48

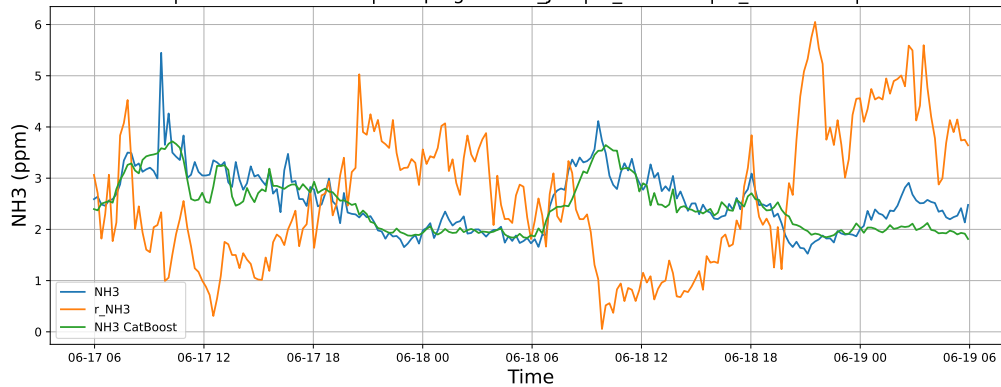
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

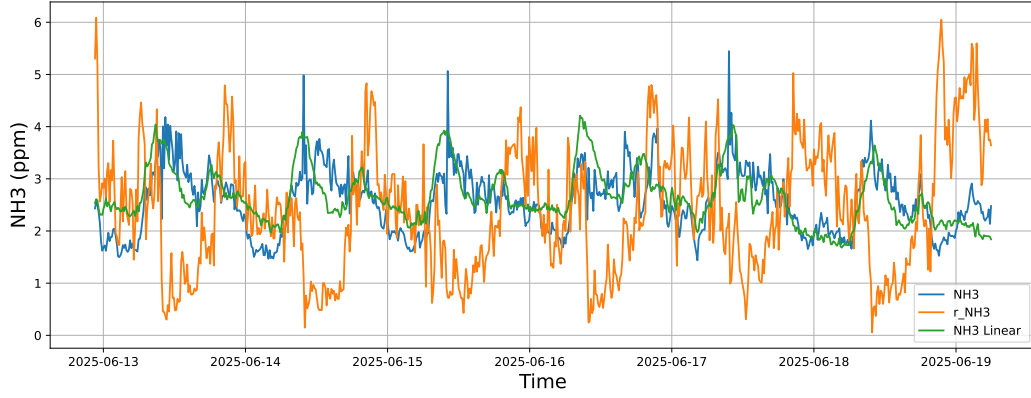


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

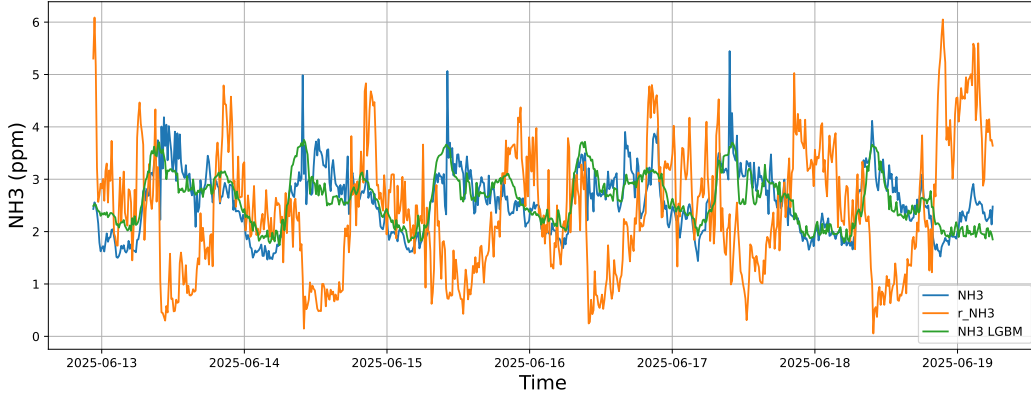


4.1.1.3 Time window : ALL

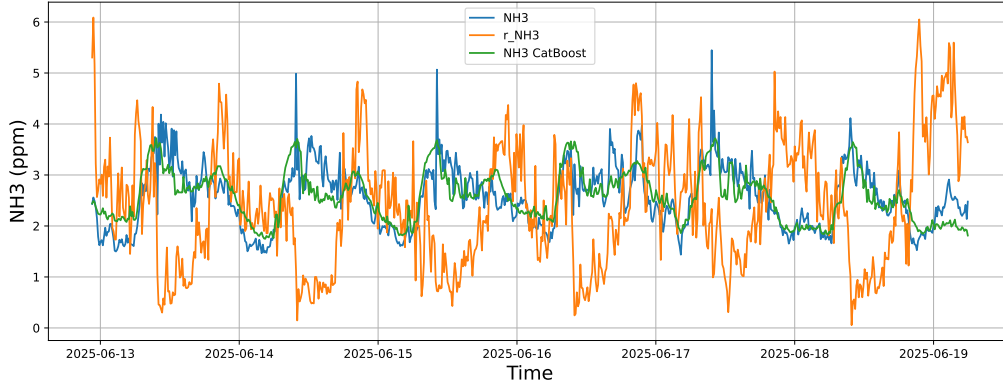
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



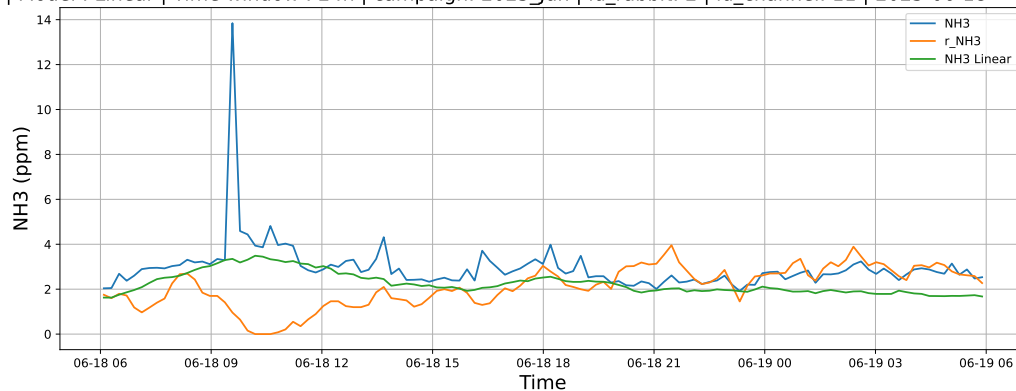
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



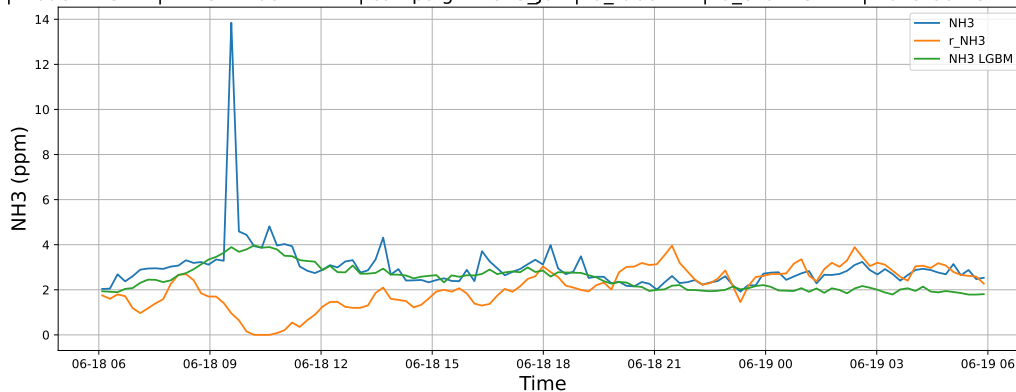
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

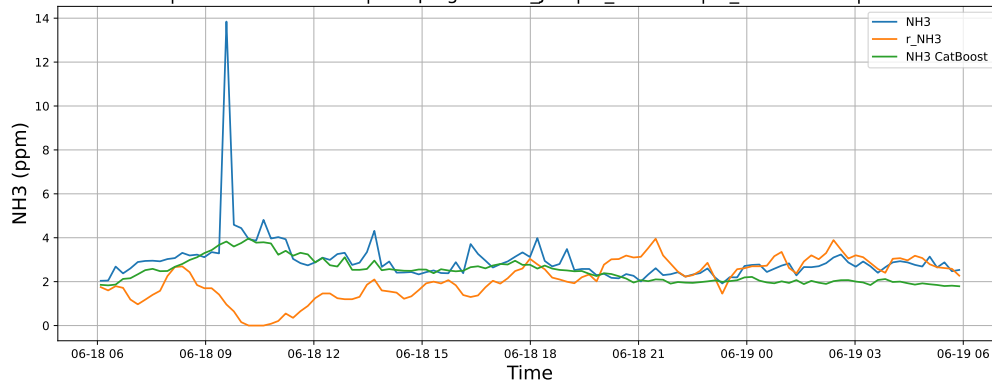
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

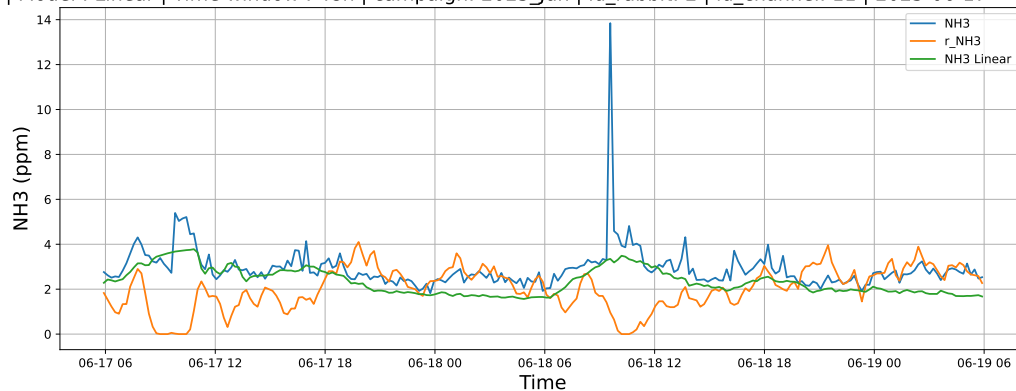


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

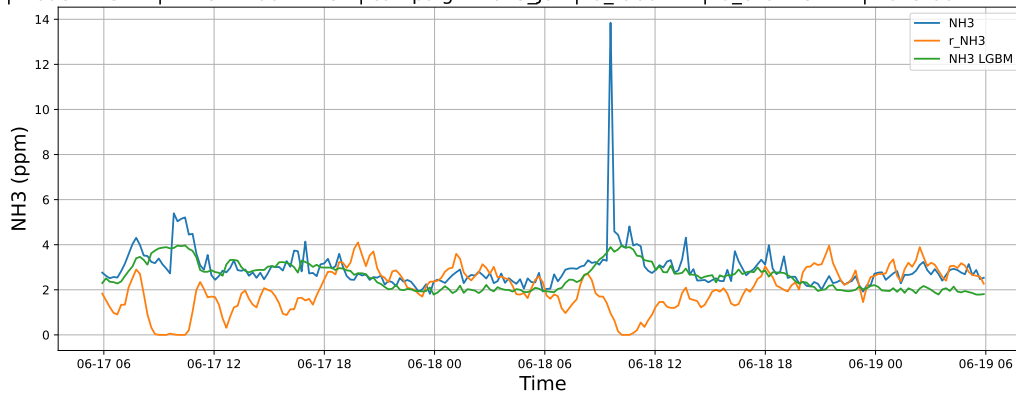


4.1.2.2 Time window : 48

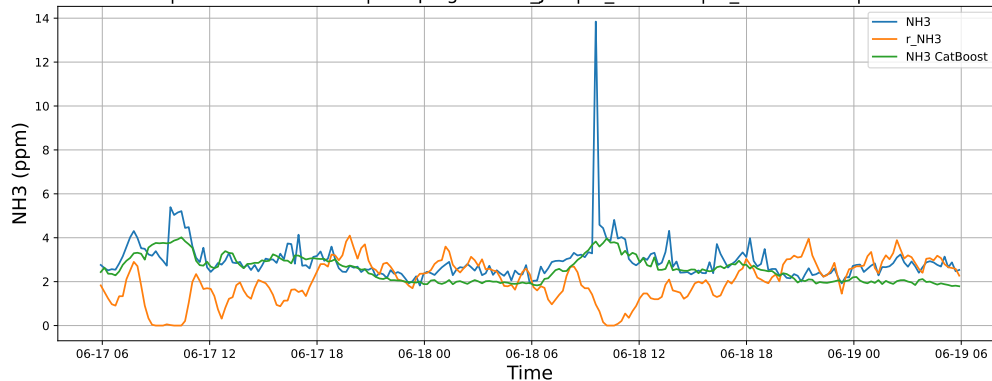
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

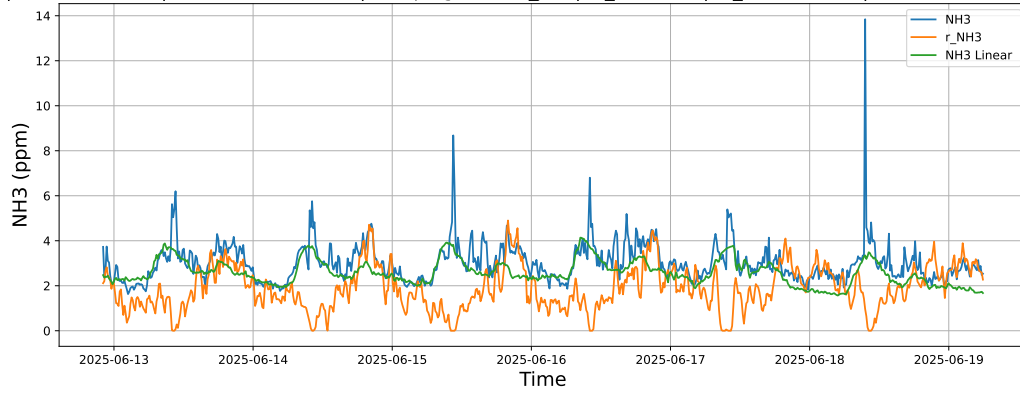


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

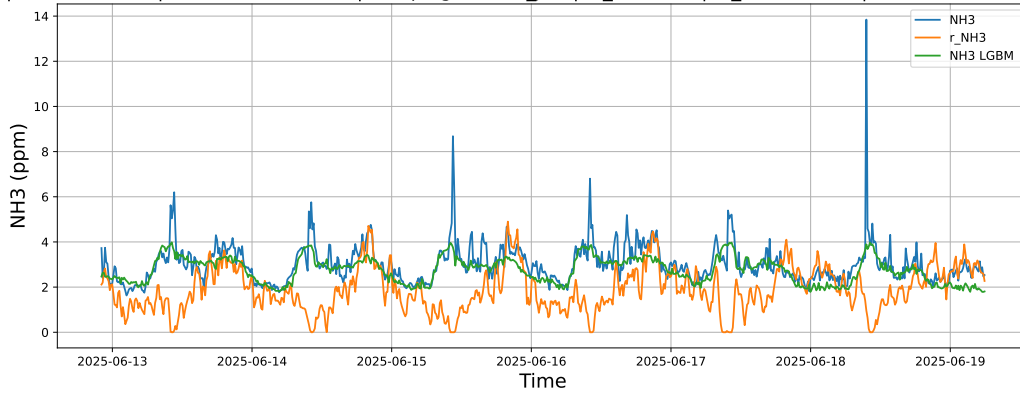


4.1.2.3 Time window : ALL

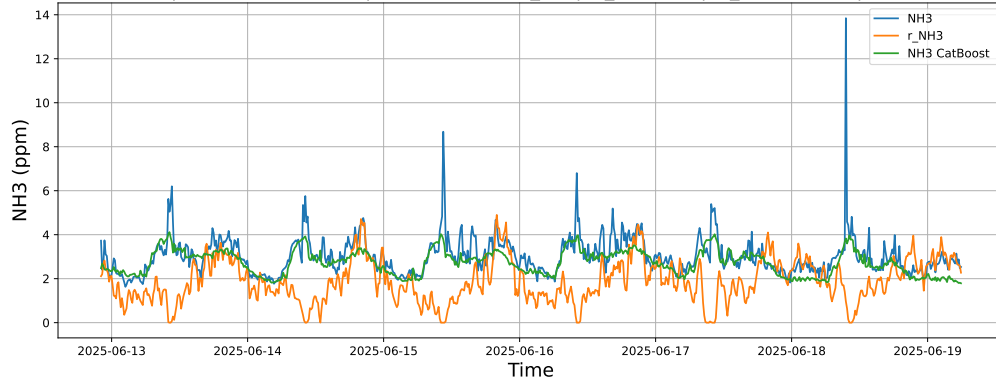
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



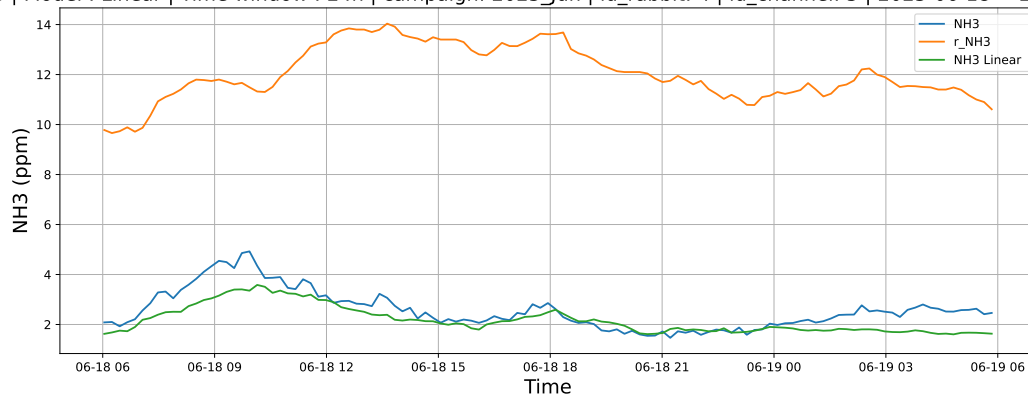
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



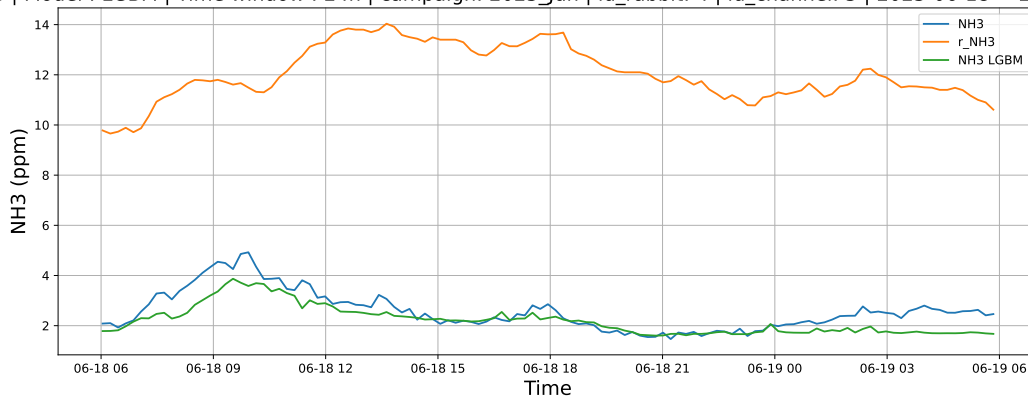
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

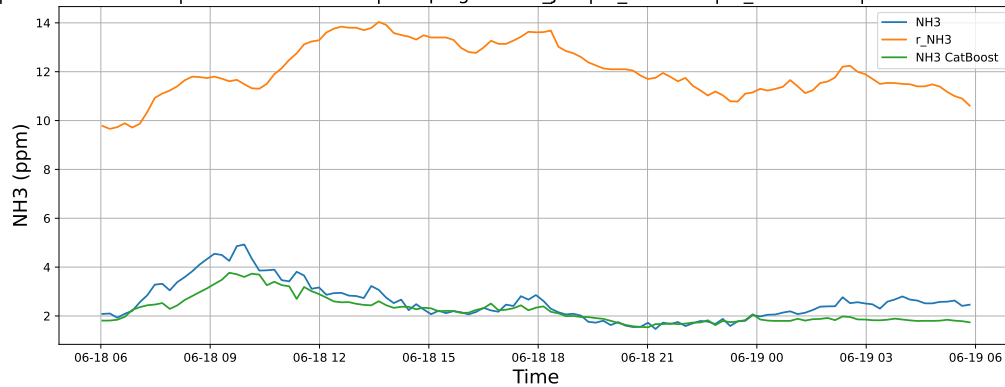
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

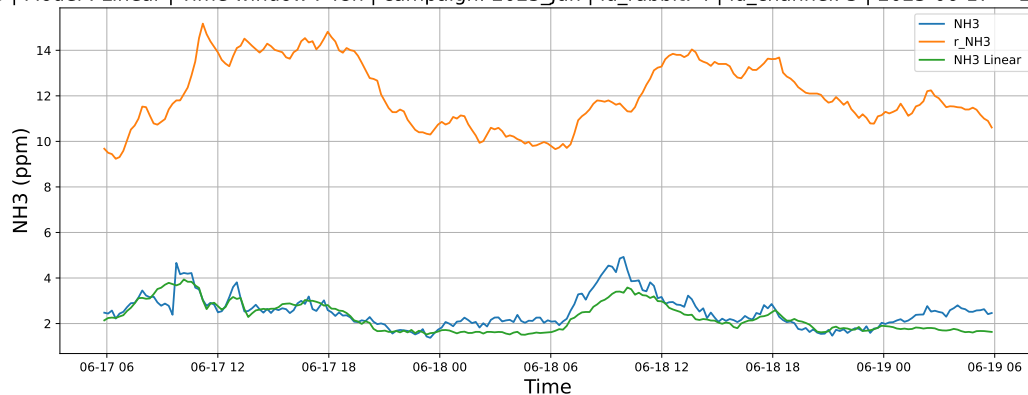


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

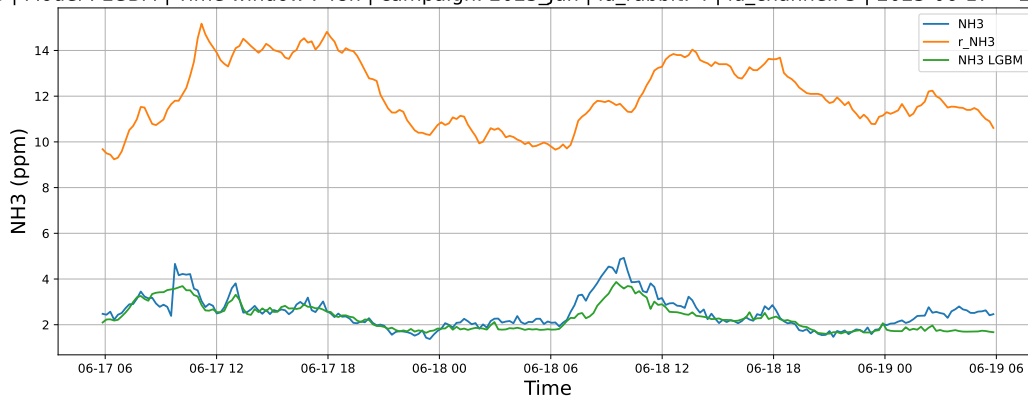


4.1.3.2 Time window : 48

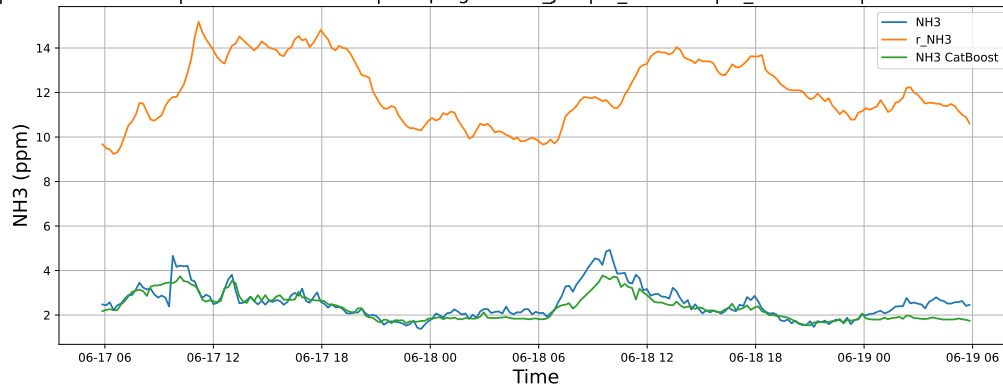
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

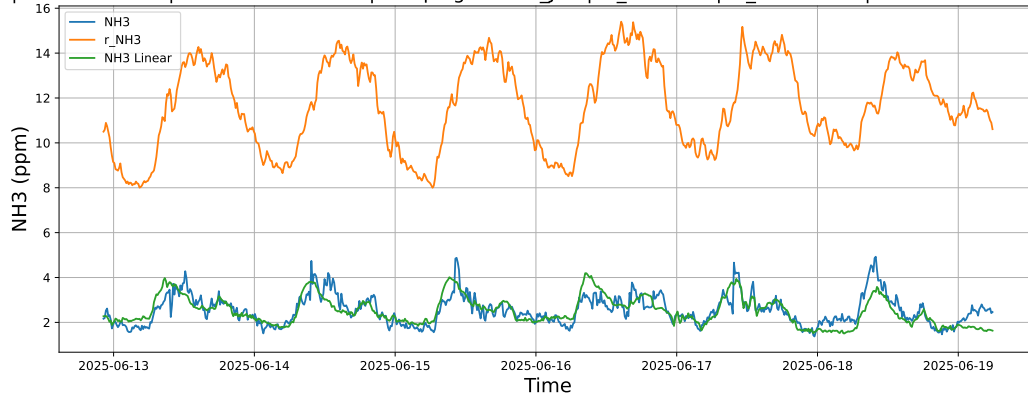


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

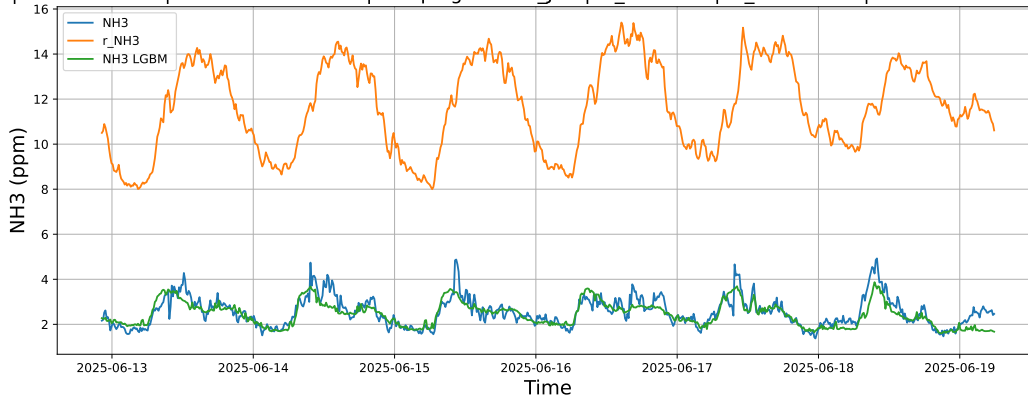


4.1.3.3 Time window : ALL

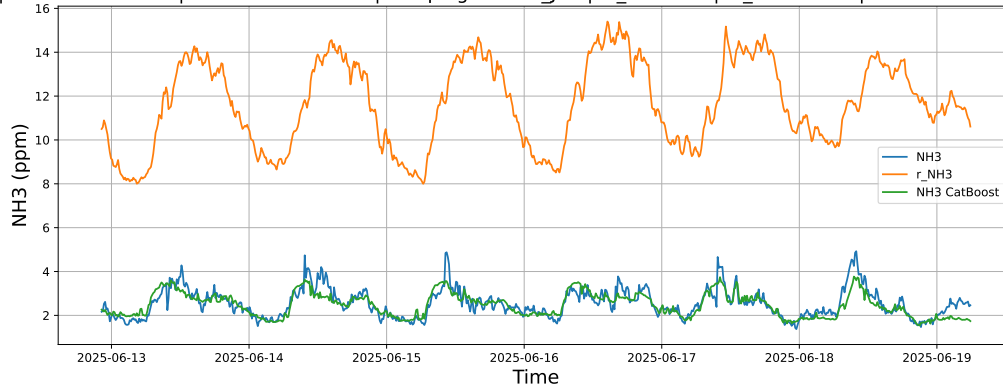
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



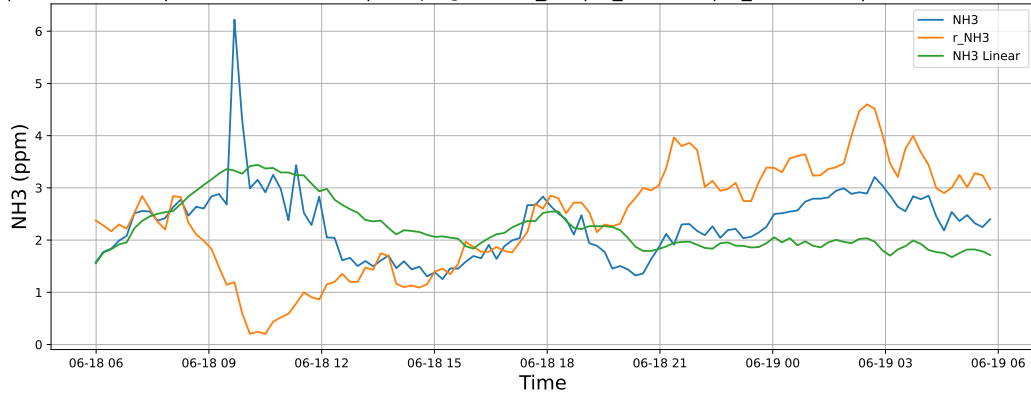
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



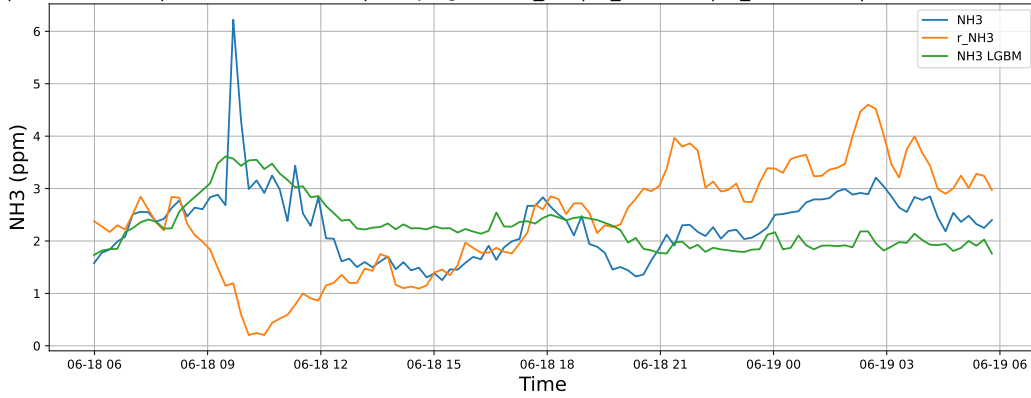
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

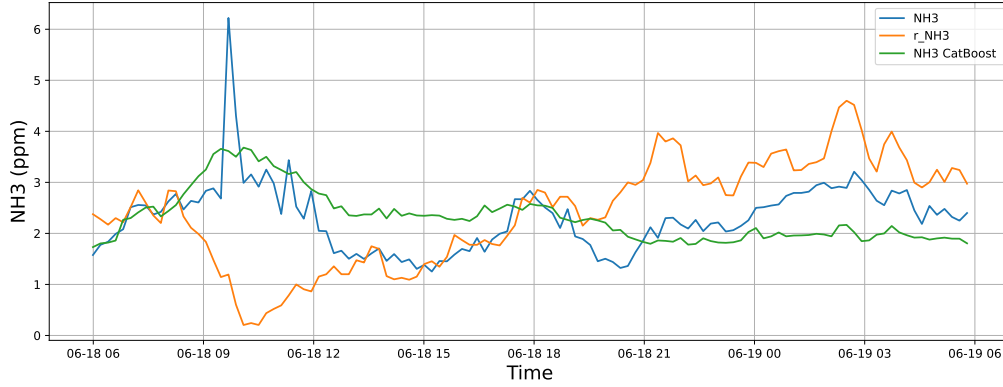
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

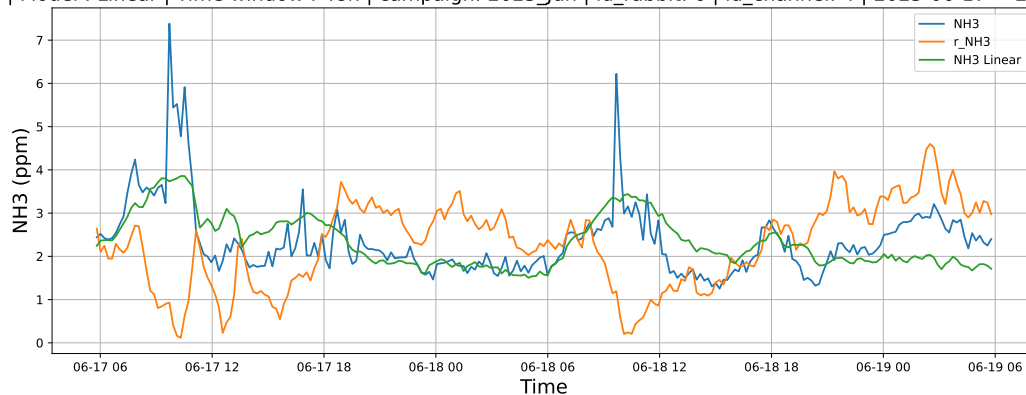


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

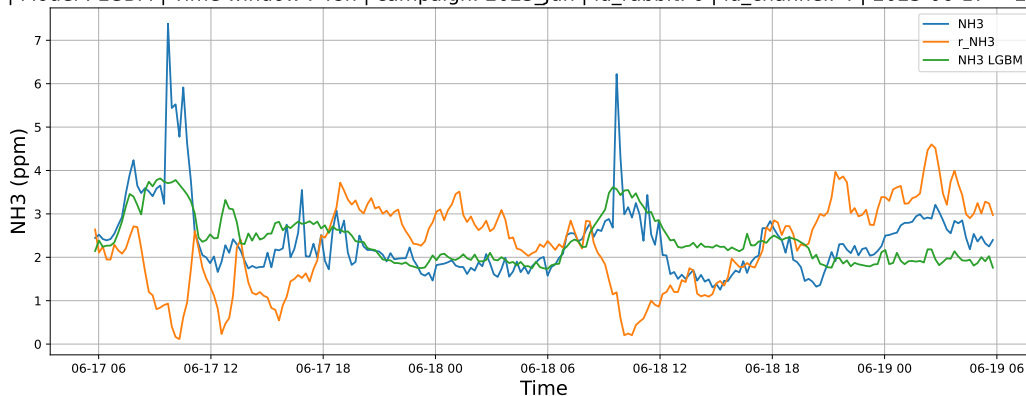


4.1.4.2 Time window : 48

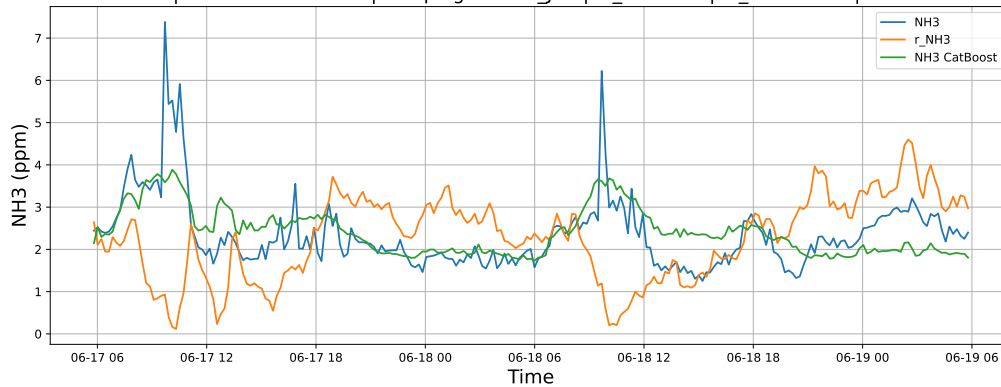
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

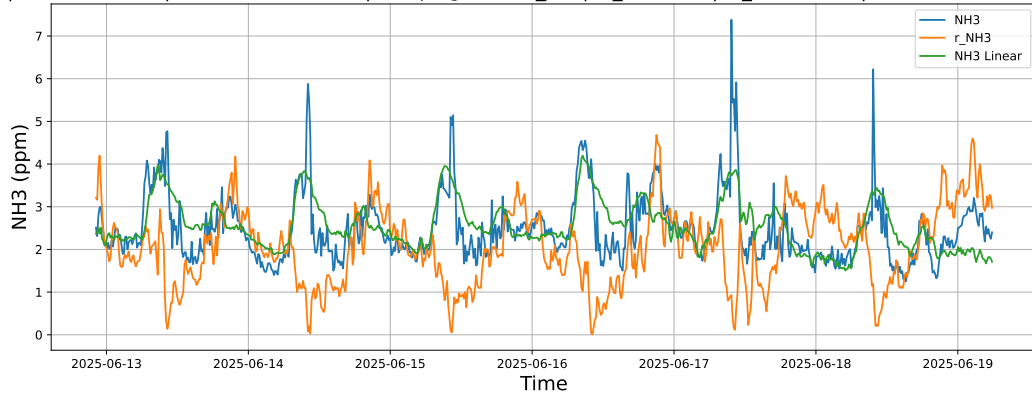


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

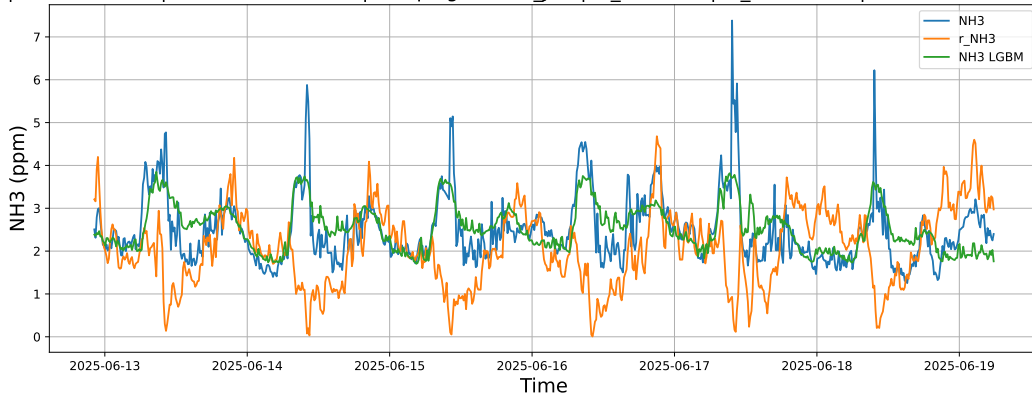


4.1.4.3 Time window : ALL

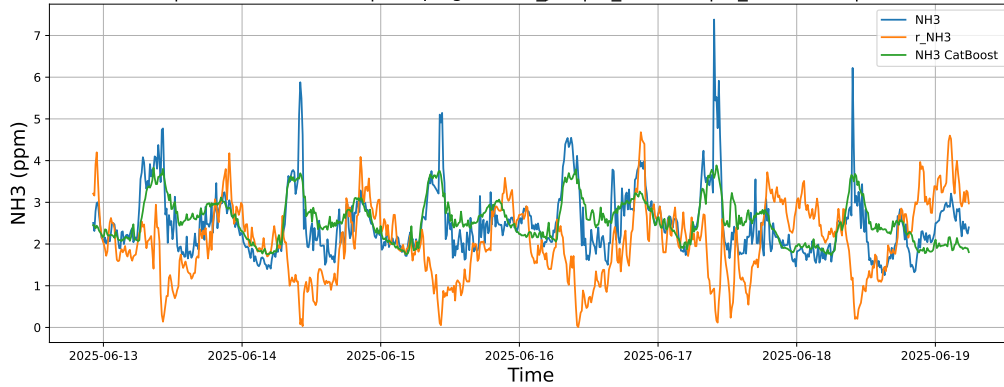
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



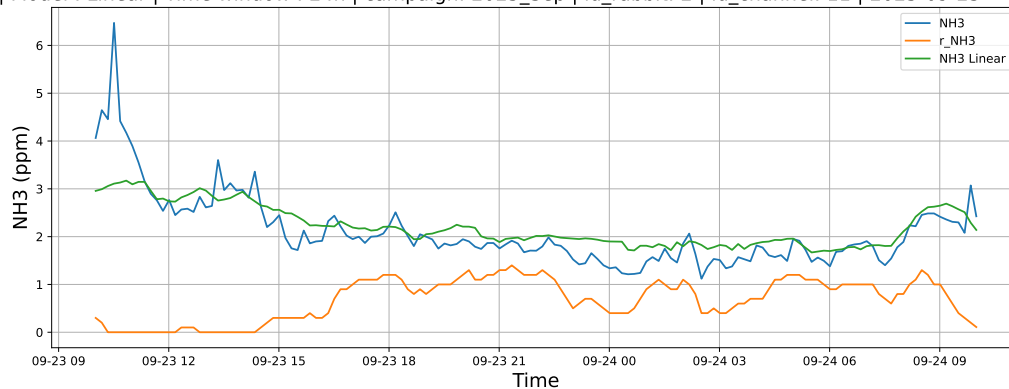
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



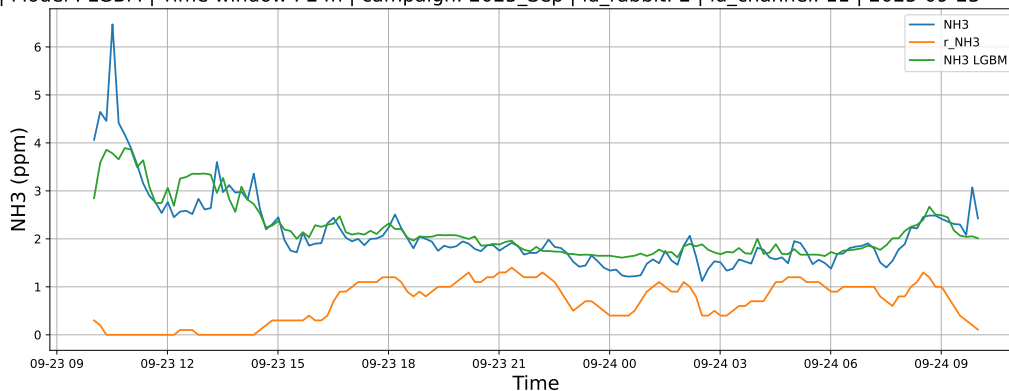
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

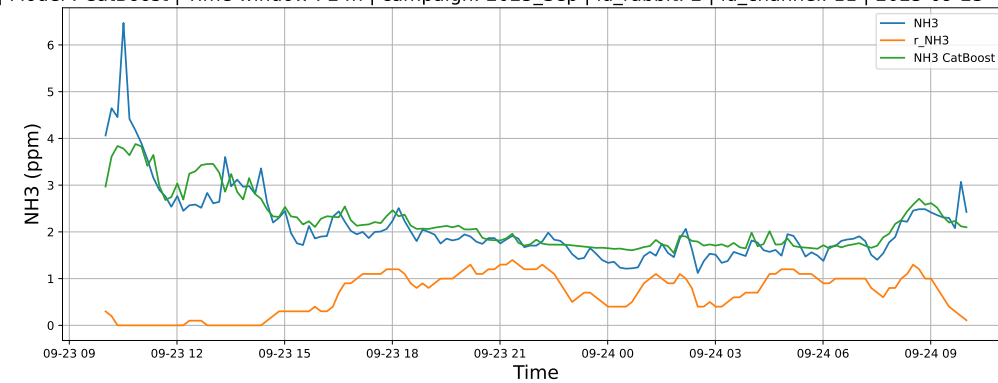
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

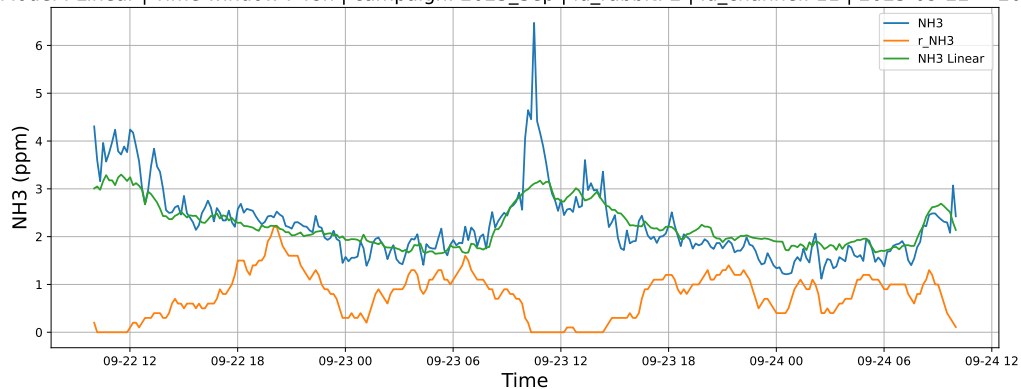


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

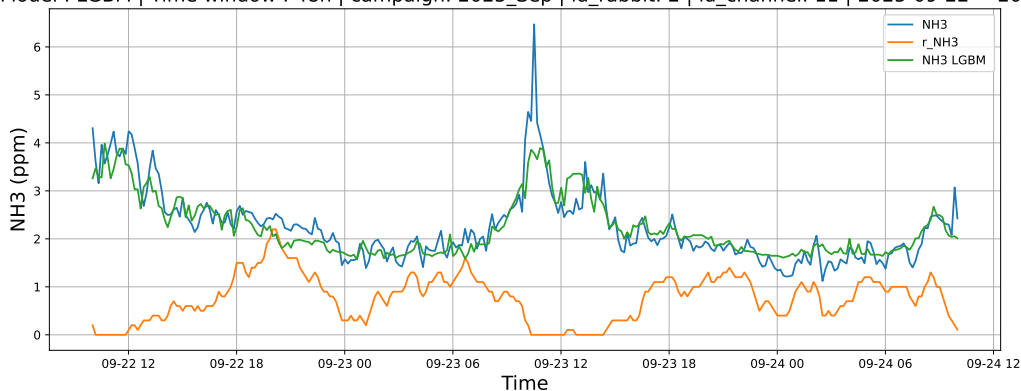


4.1.5.2 Time window : 48

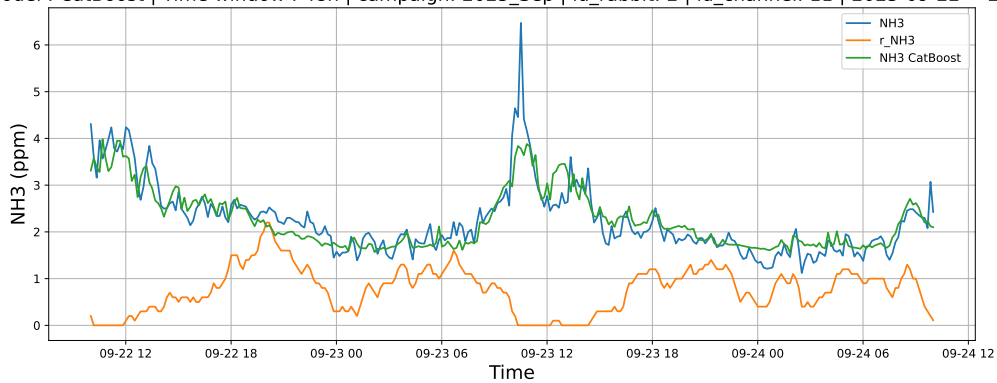
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

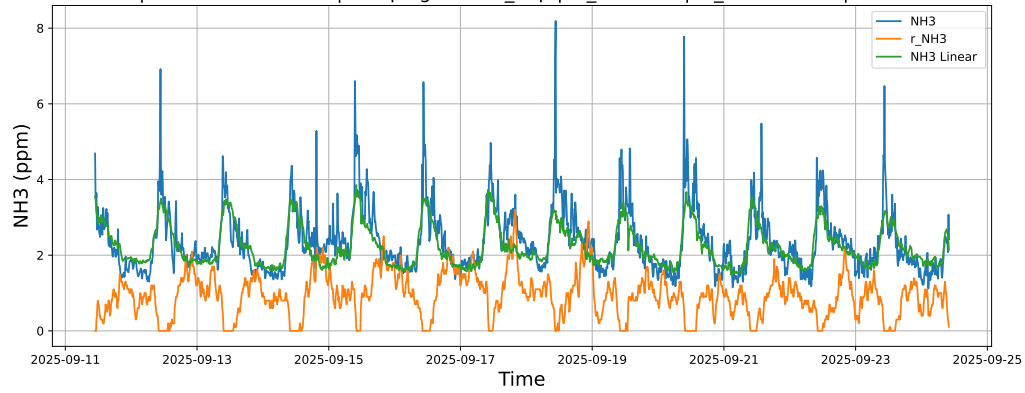


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

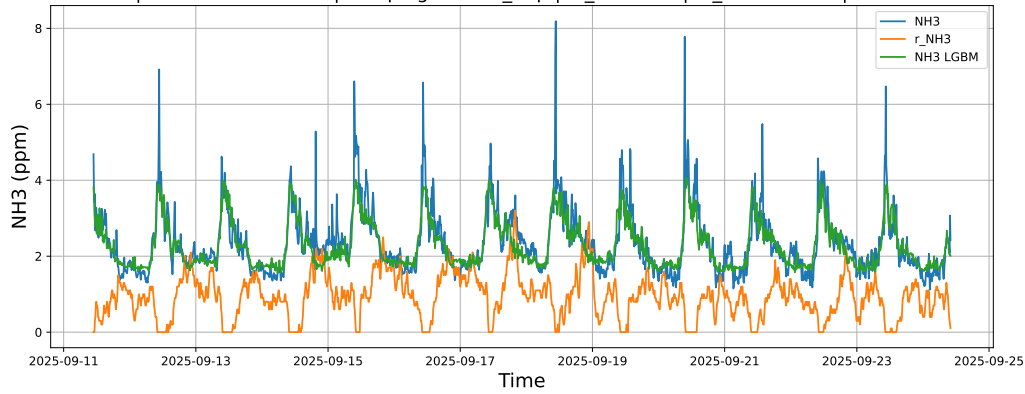


4.1.5.3 Time window : ALL

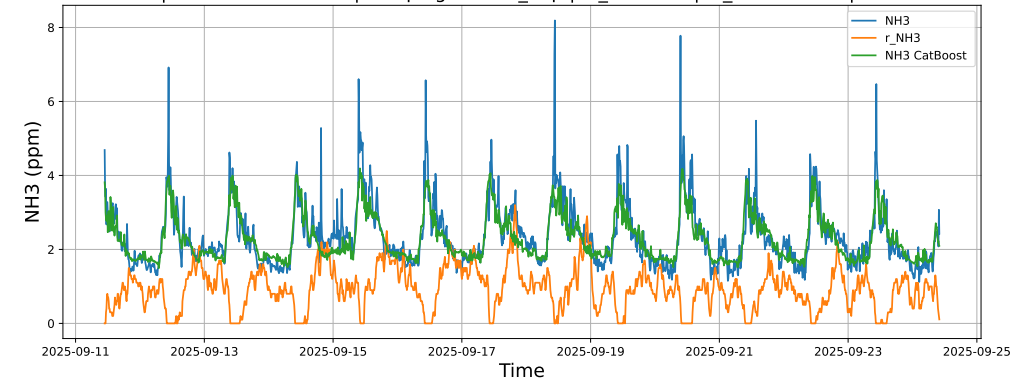
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



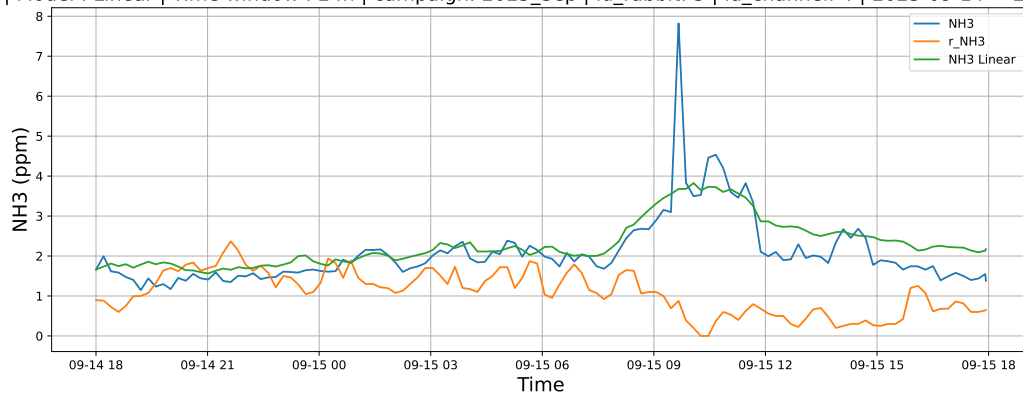
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



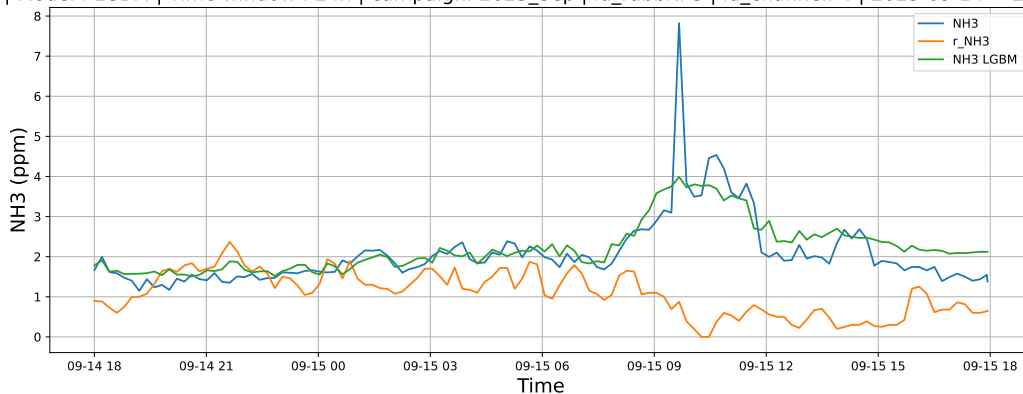
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

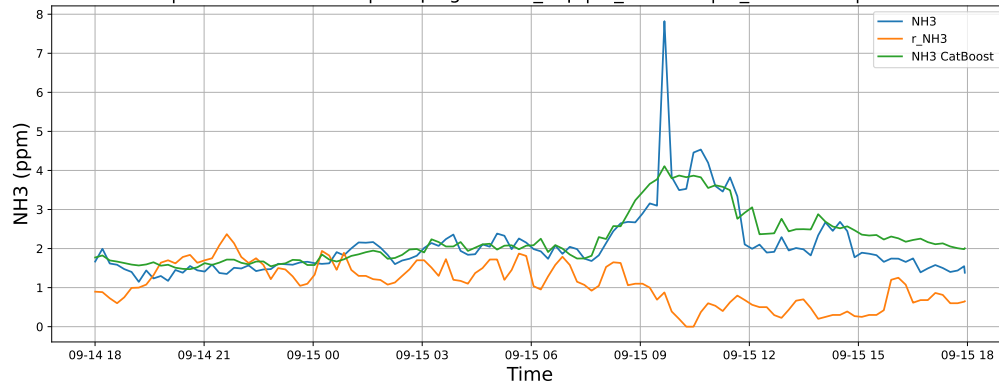
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

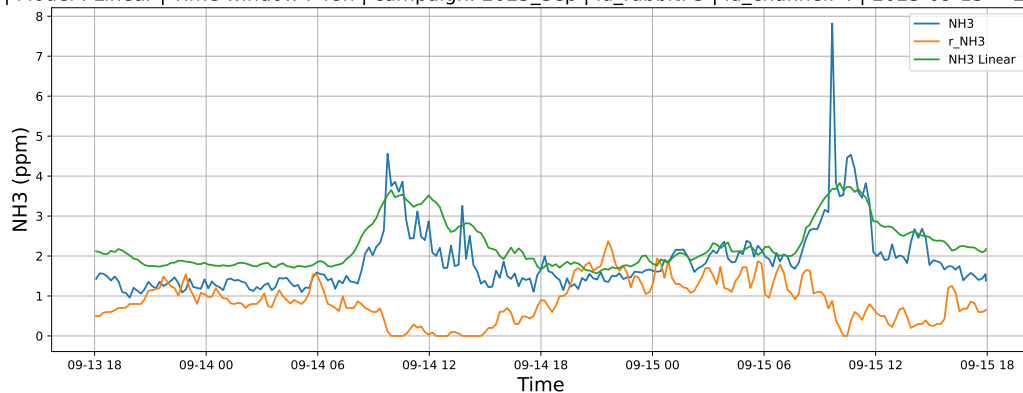


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

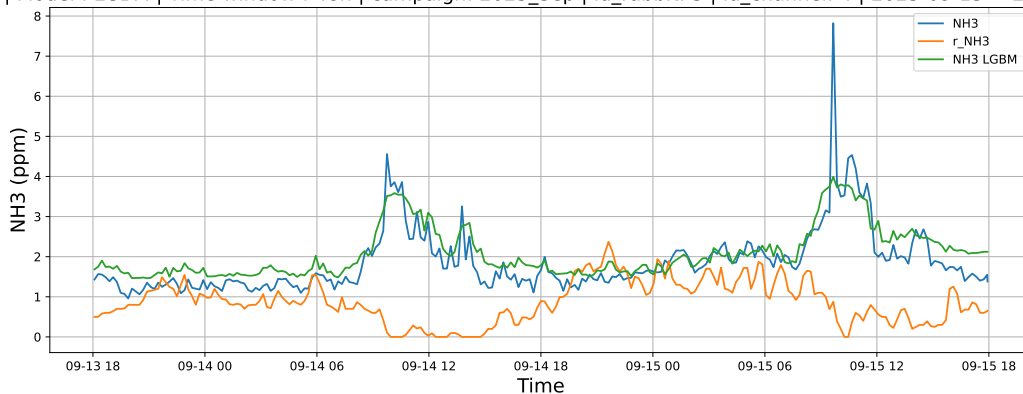


4.1.6.2 Time window : 48

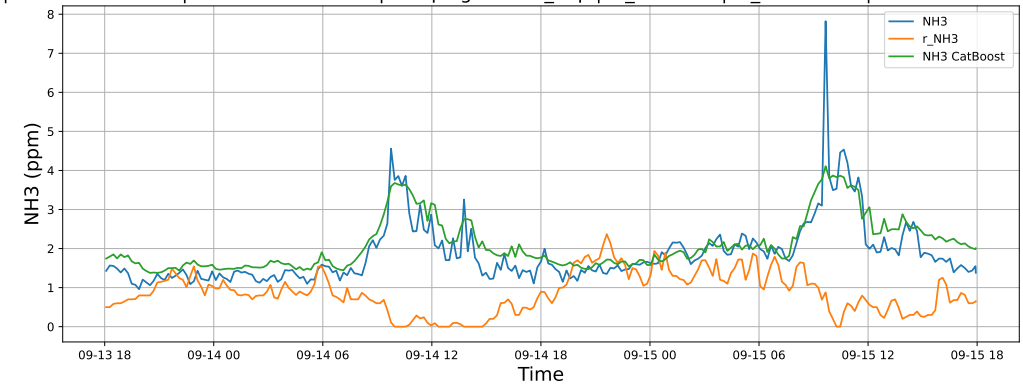
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

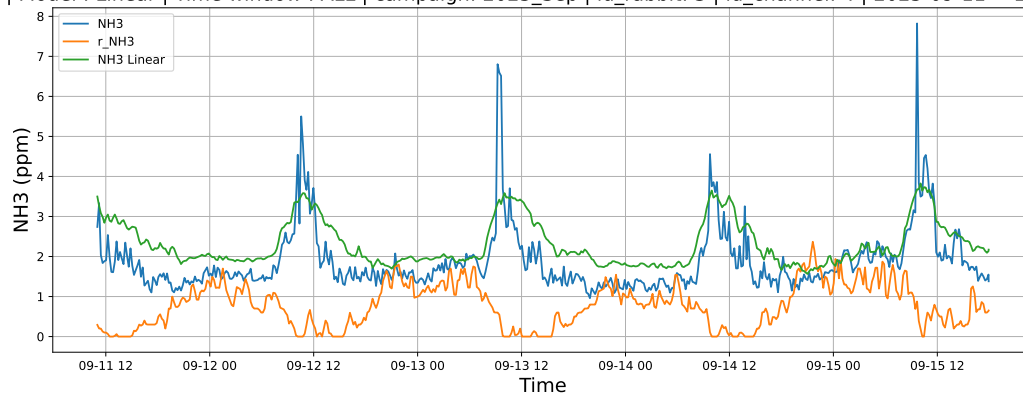


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

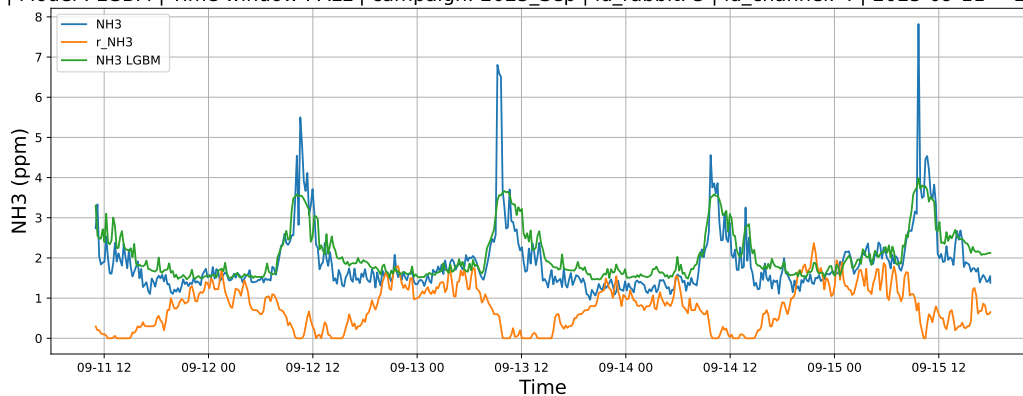


4.1.6.3 Time window : ALL

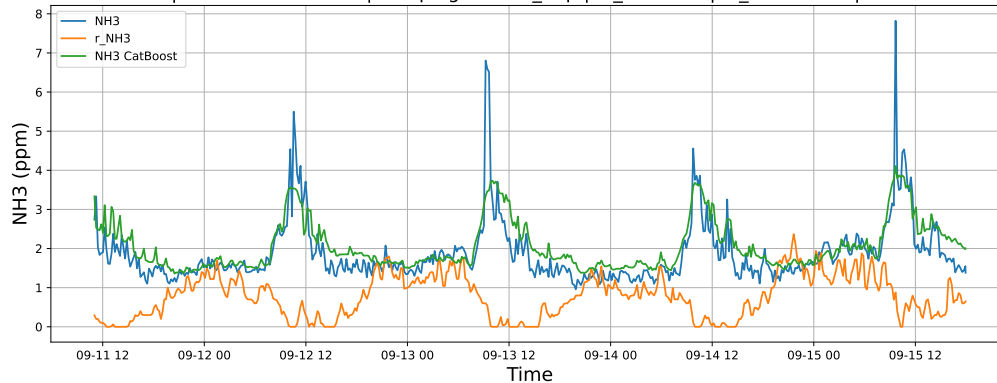
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



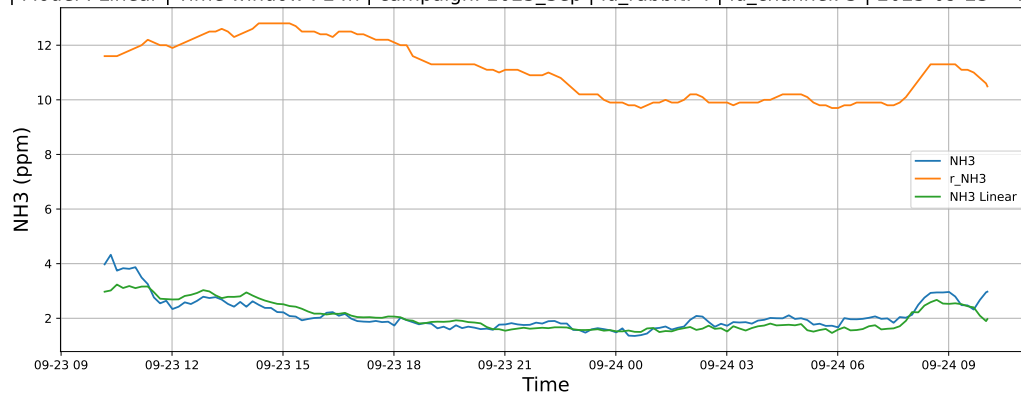
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



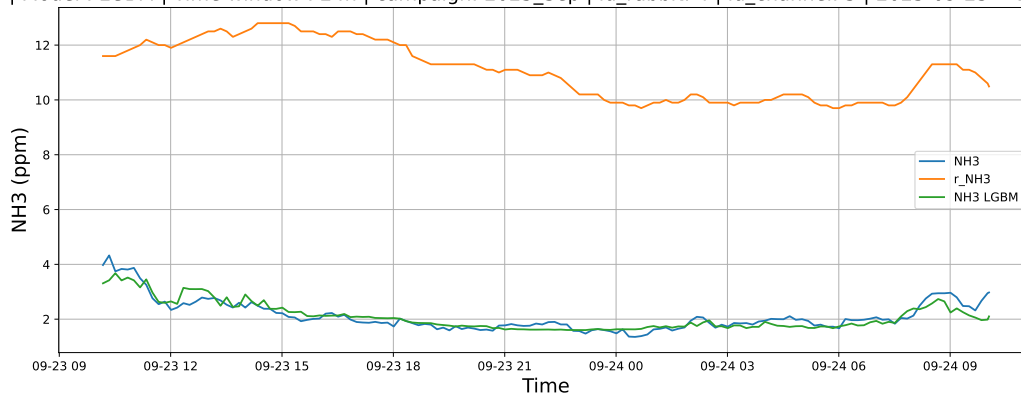
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

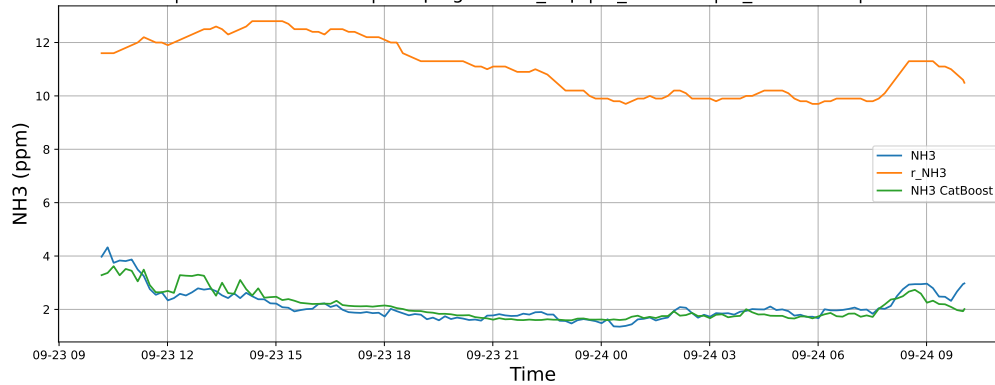
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

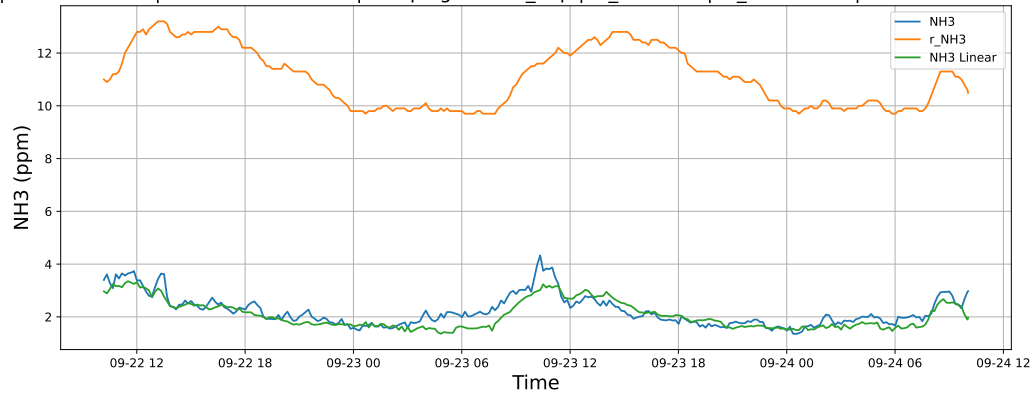


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

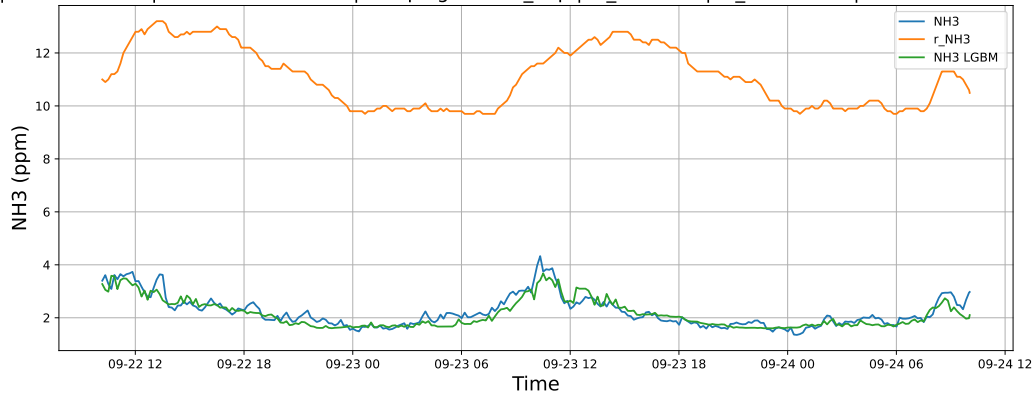


4.1.7.2 Time window : 48

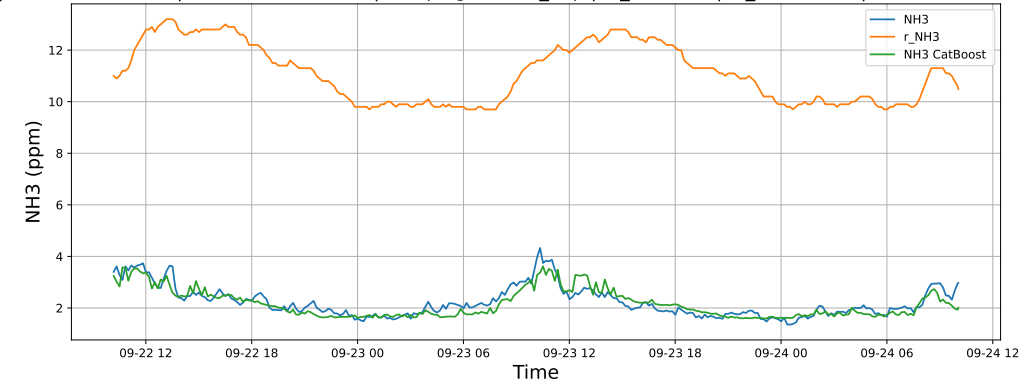
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

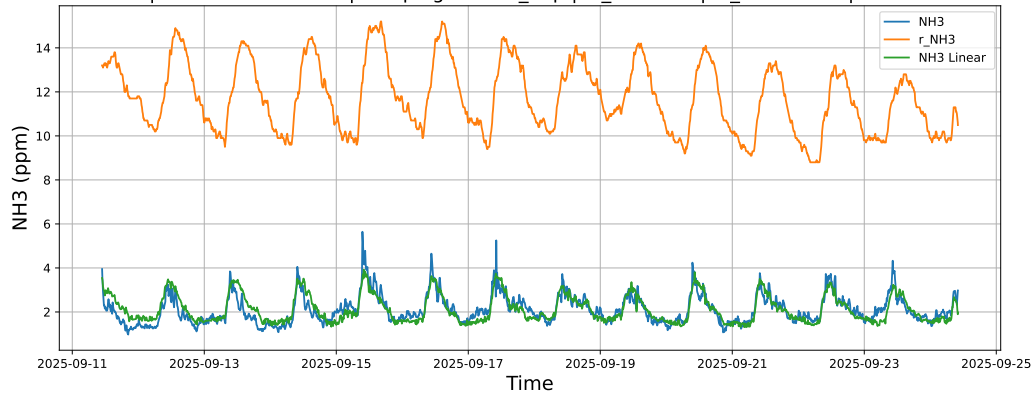


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

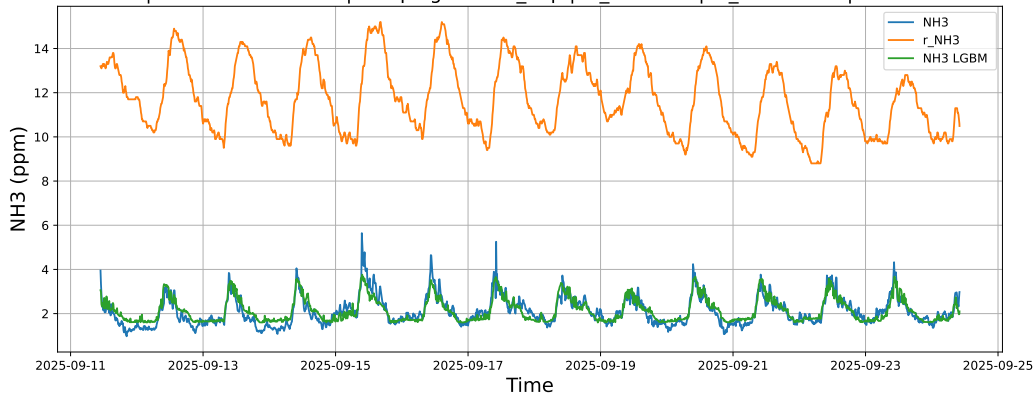


4.1.7.3 Time window : ALL

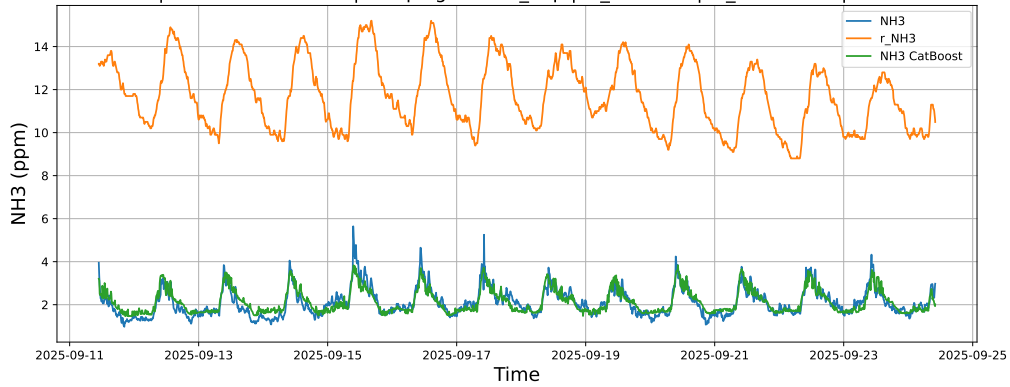
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



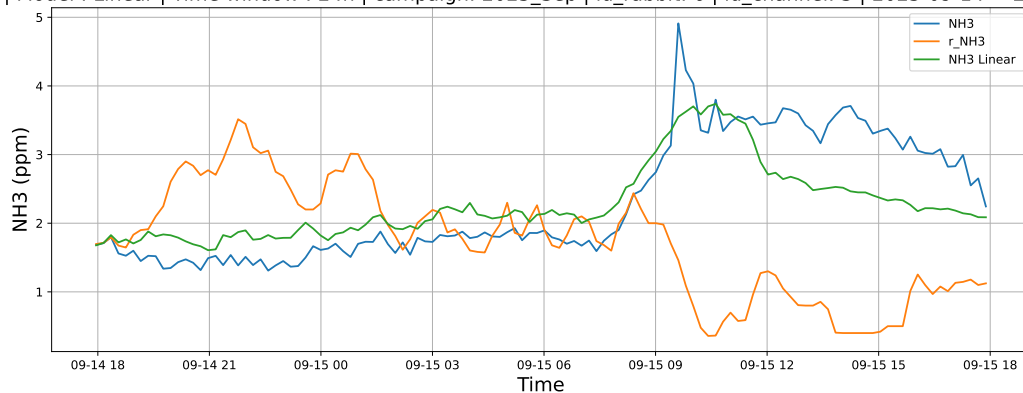
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



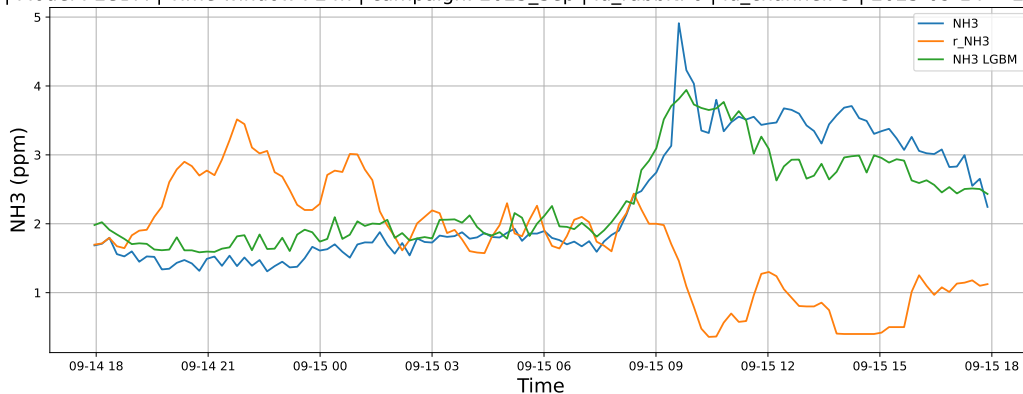
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

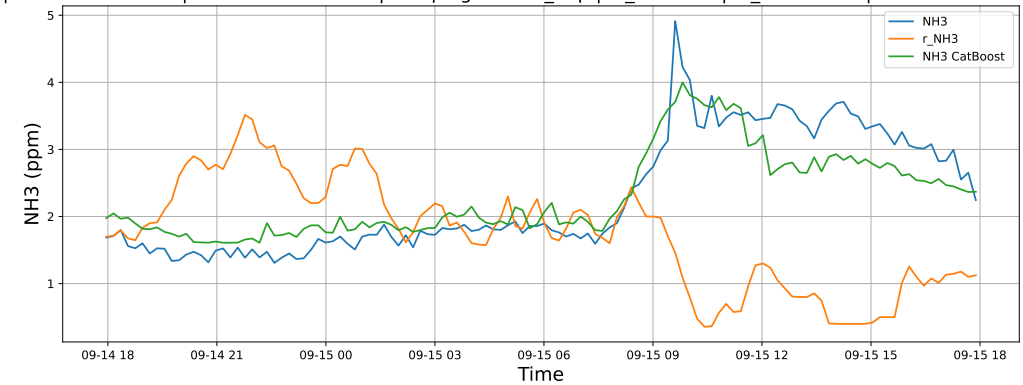
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

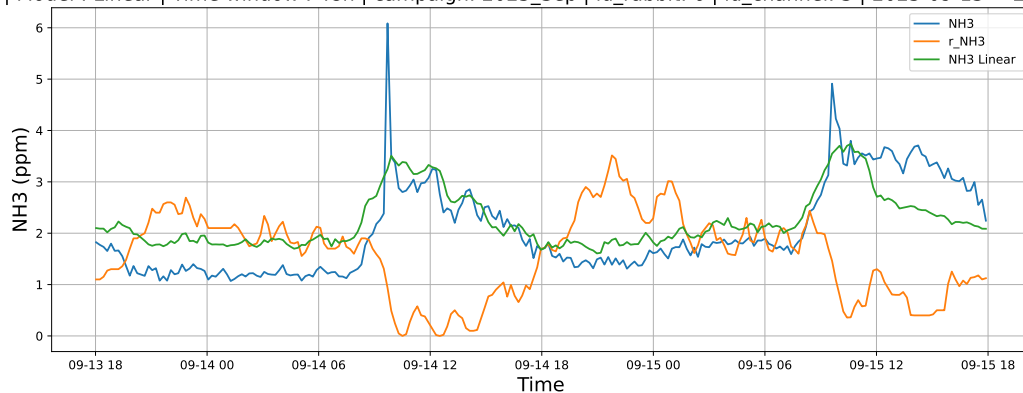


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

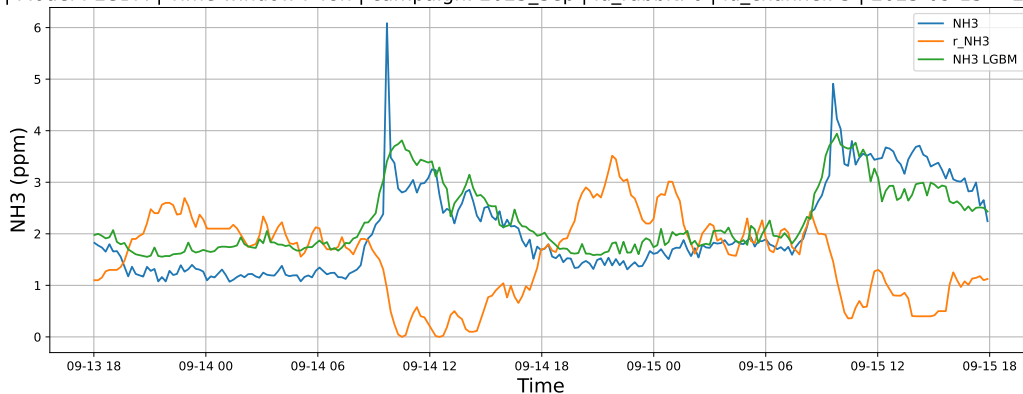


4.1.8.2 Time window : 48

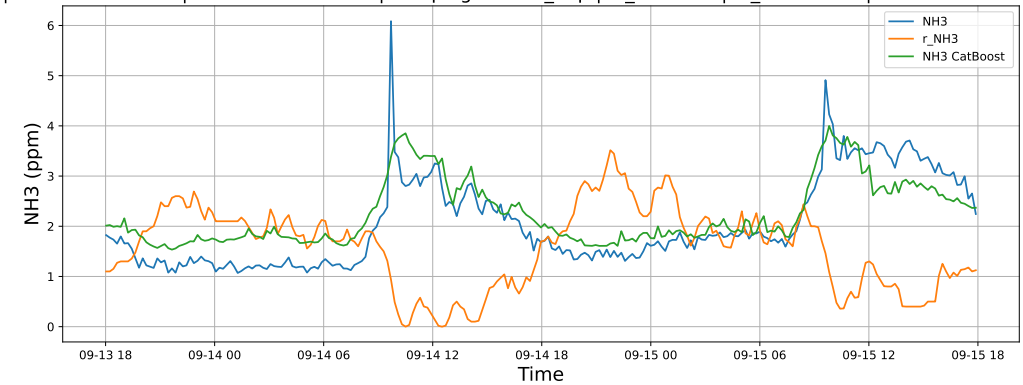
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

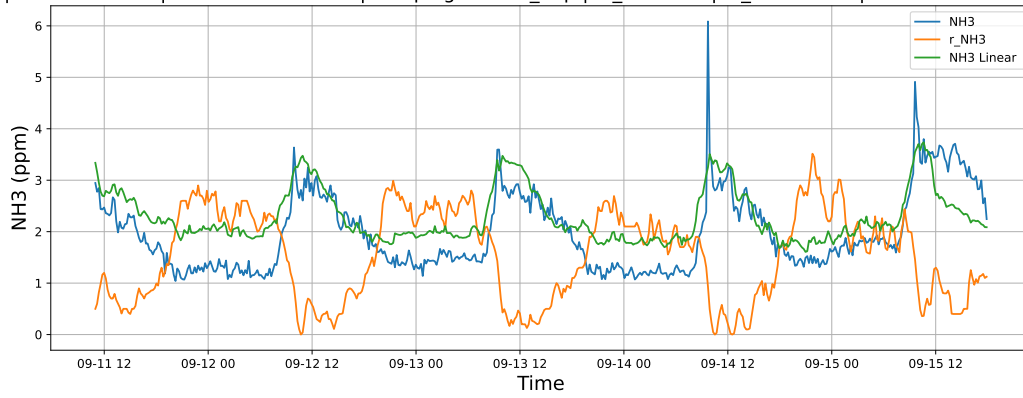


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

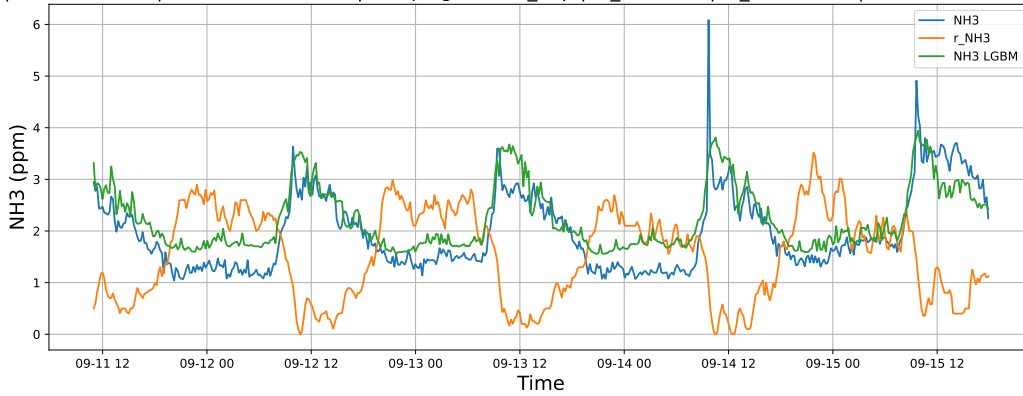


4.1.8.3 Time window : ALL

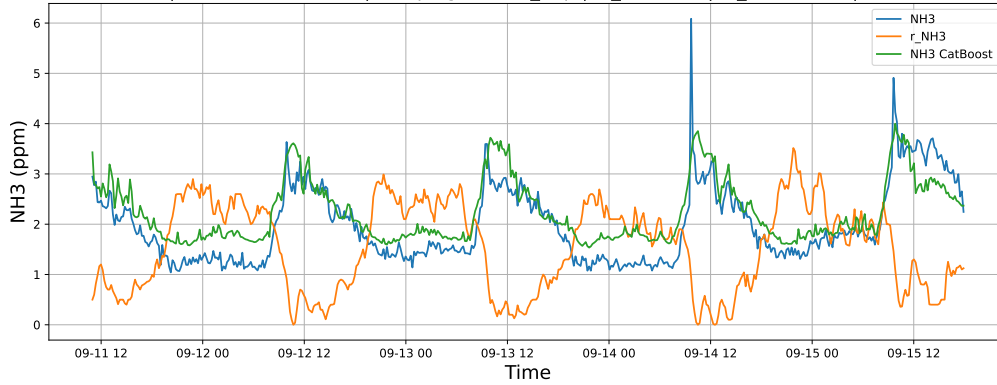
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

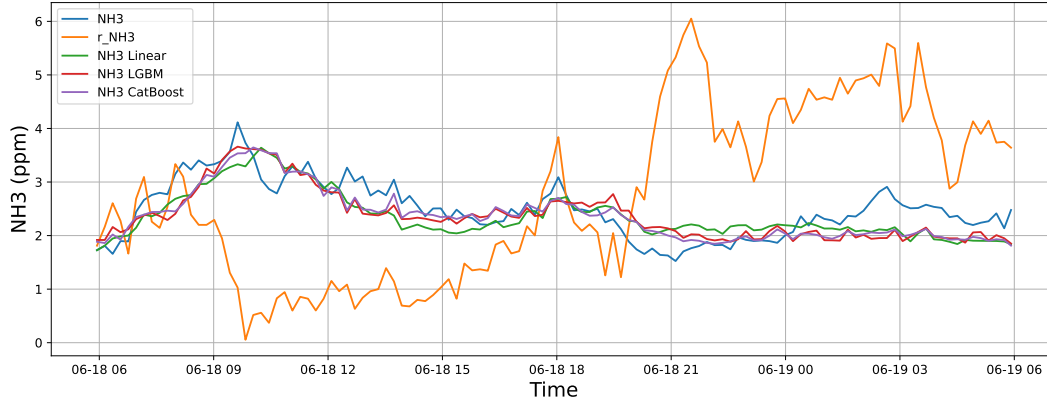


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

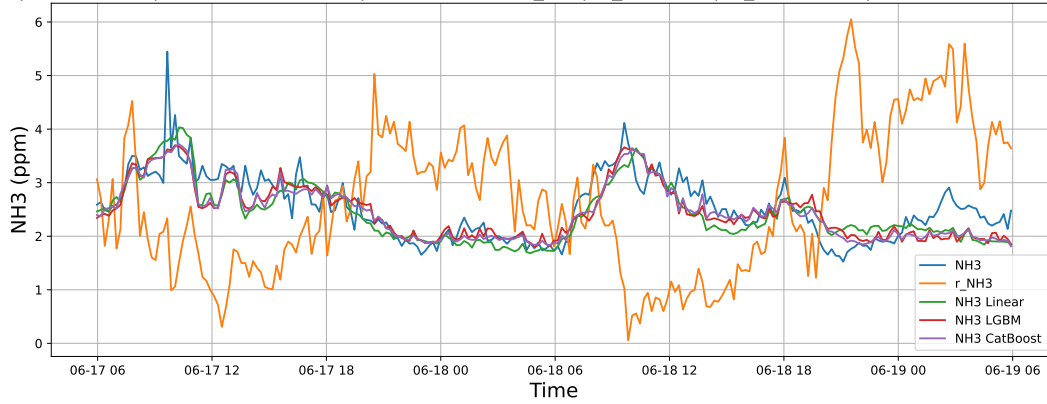
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



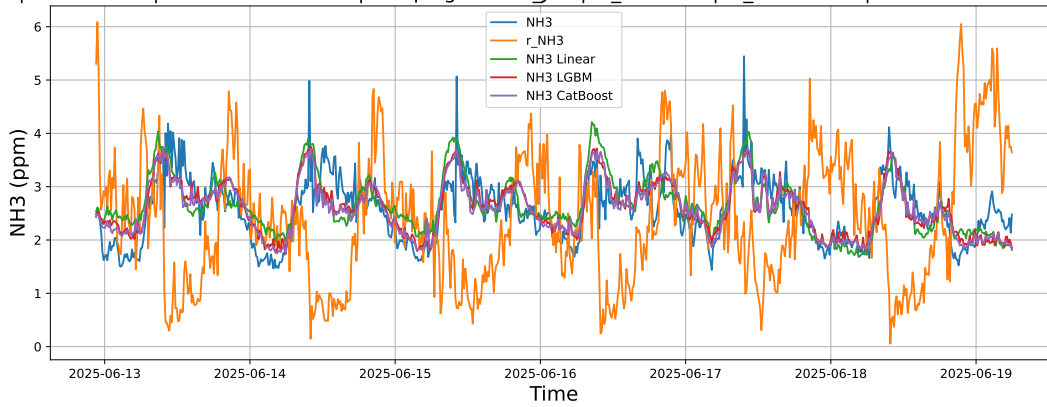
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

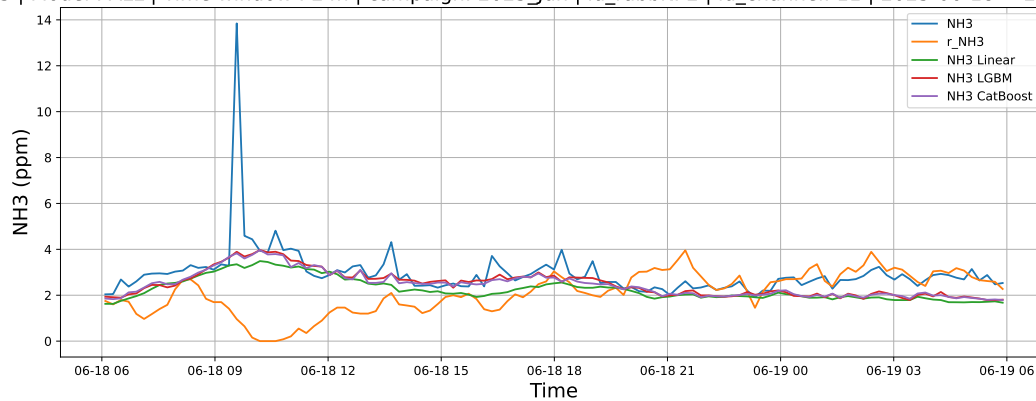
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

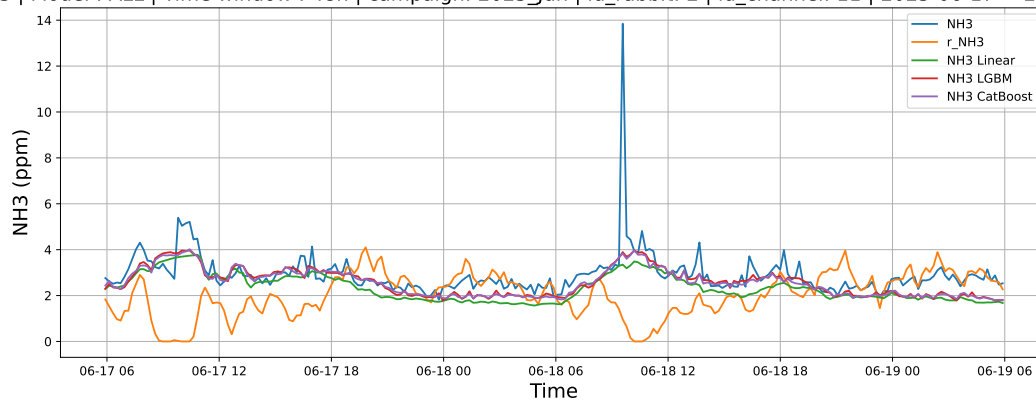
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



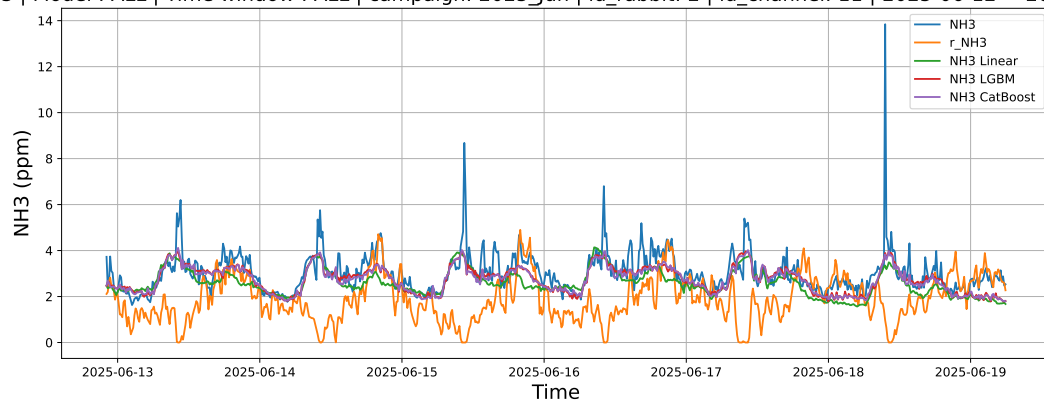
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

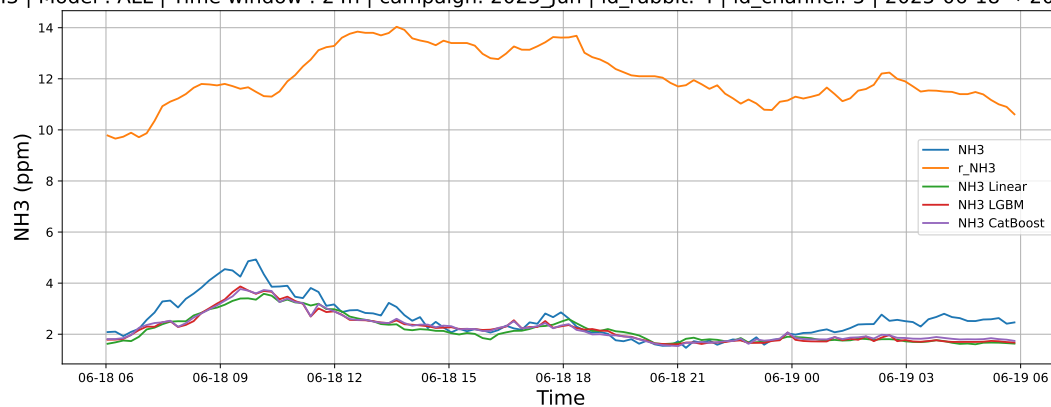
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

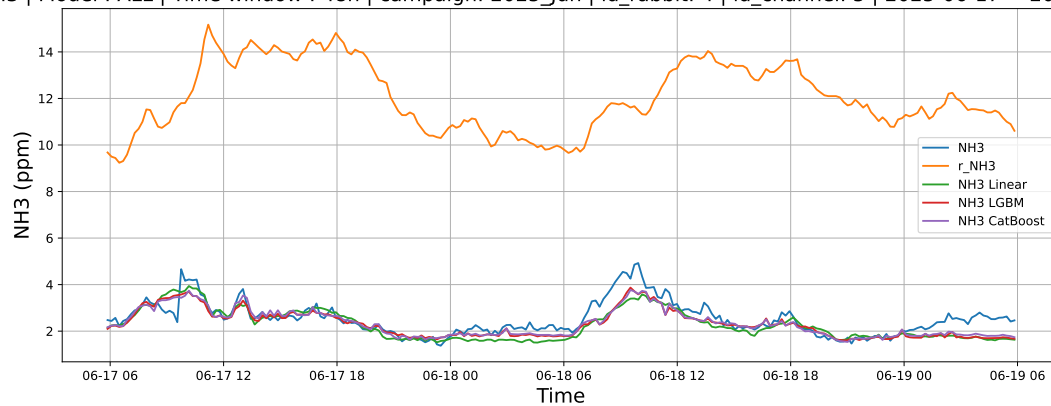
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



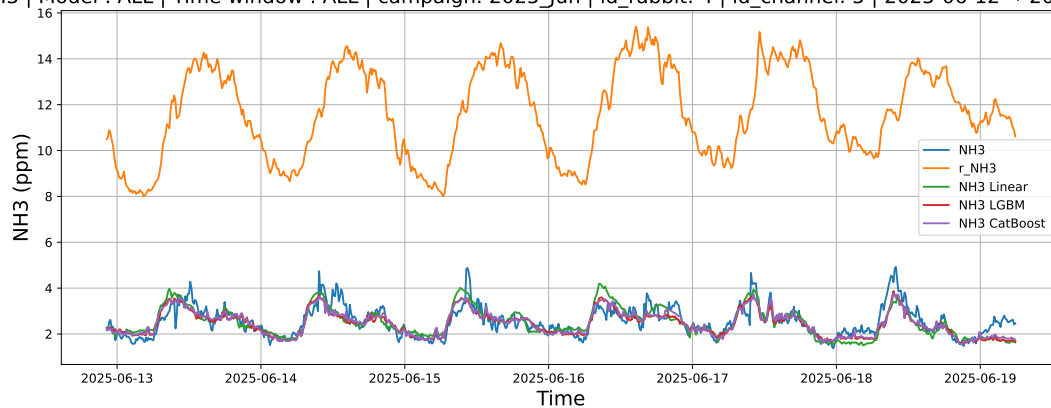
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

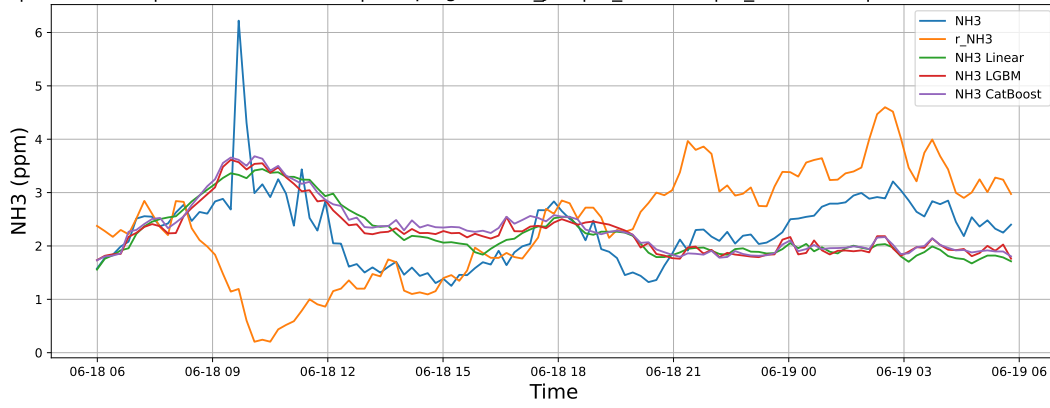
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

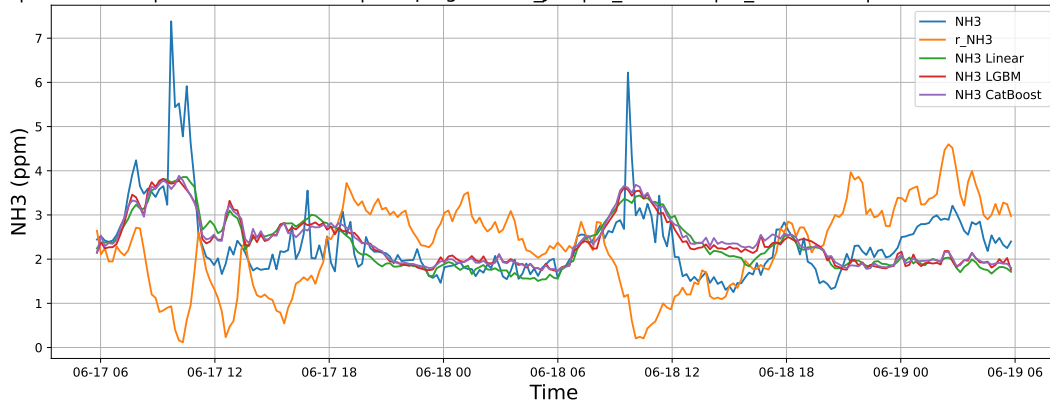
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



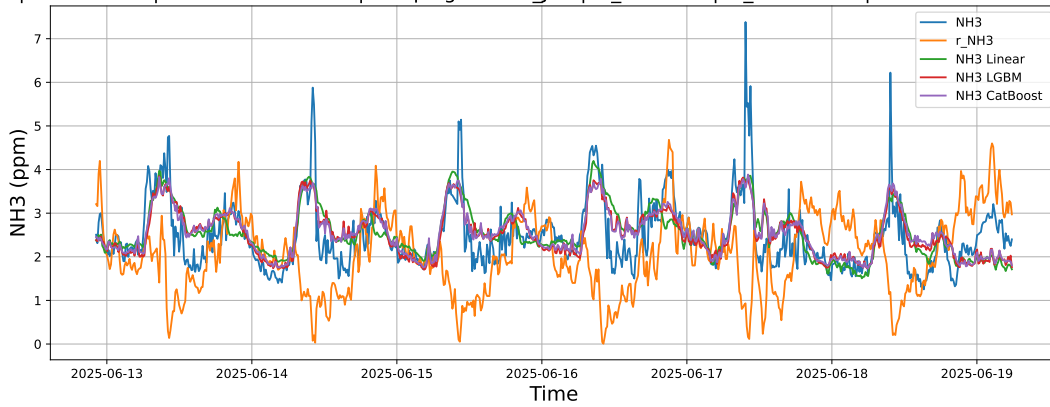
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

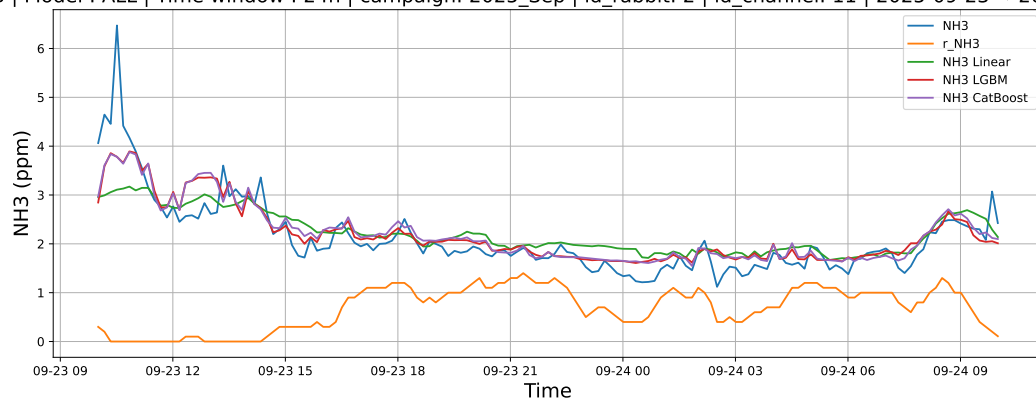
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

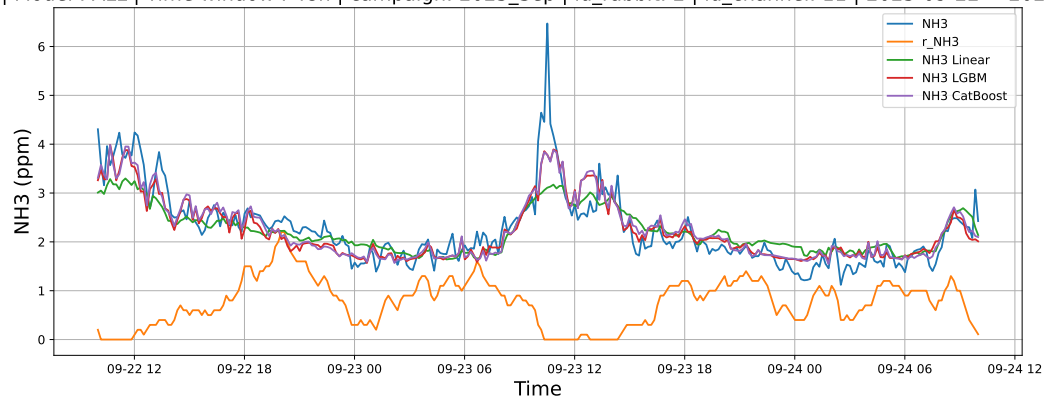
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



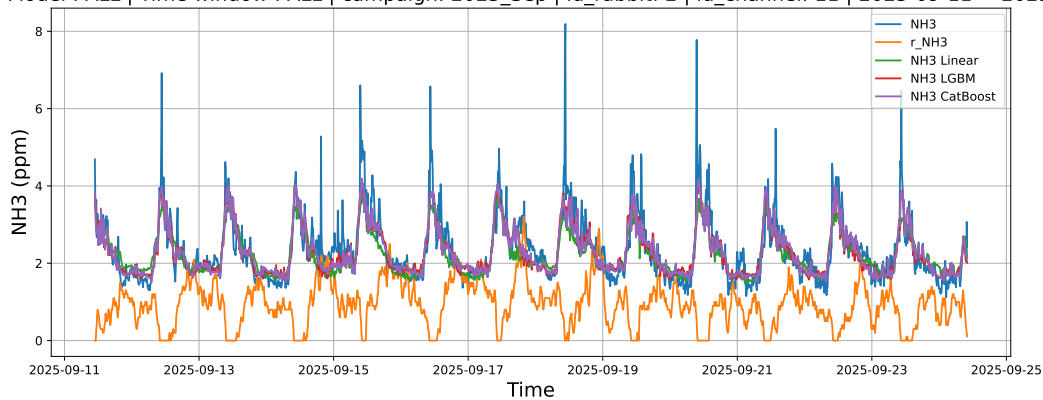
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

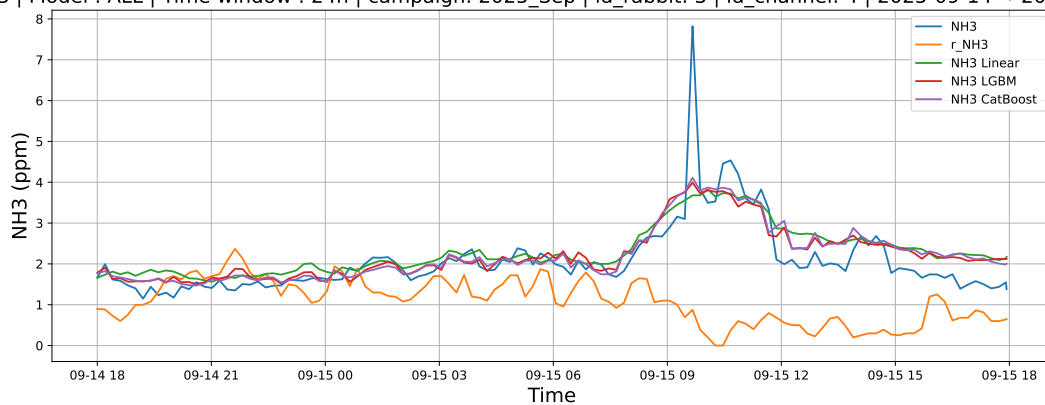
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

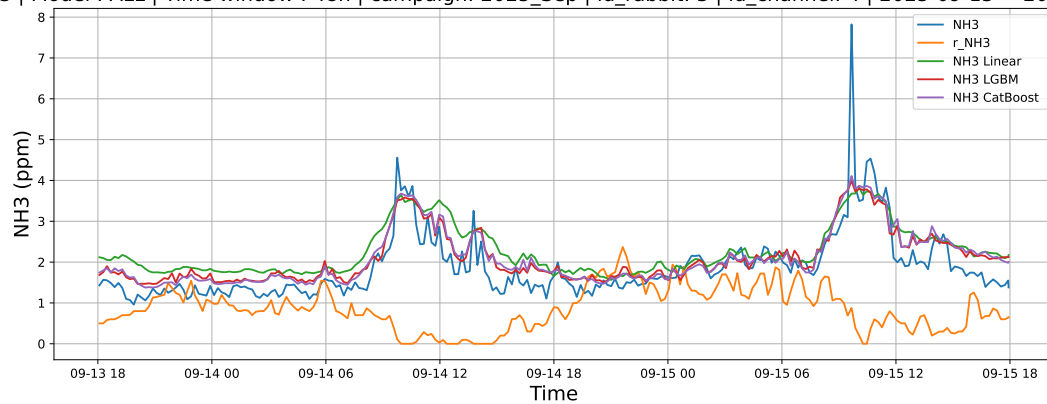
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



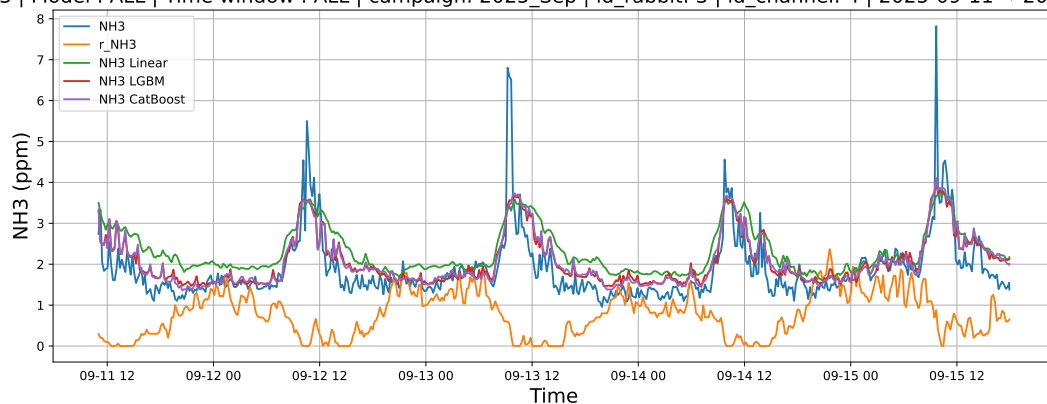
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

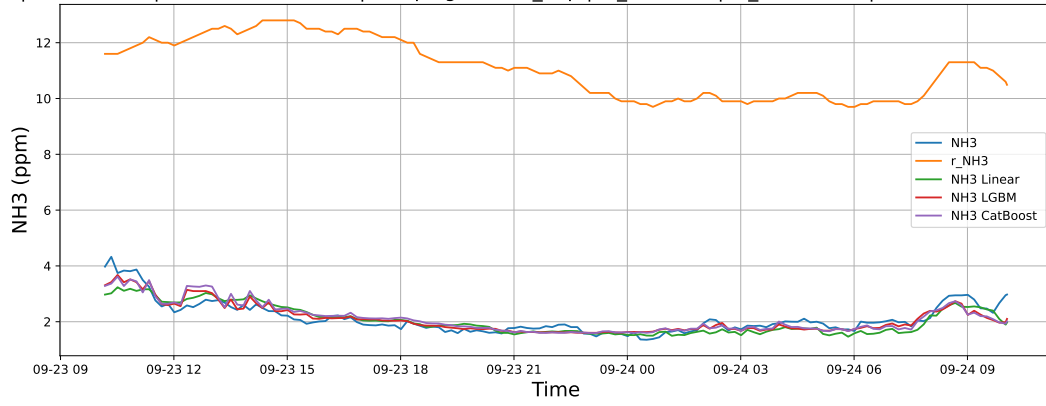
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

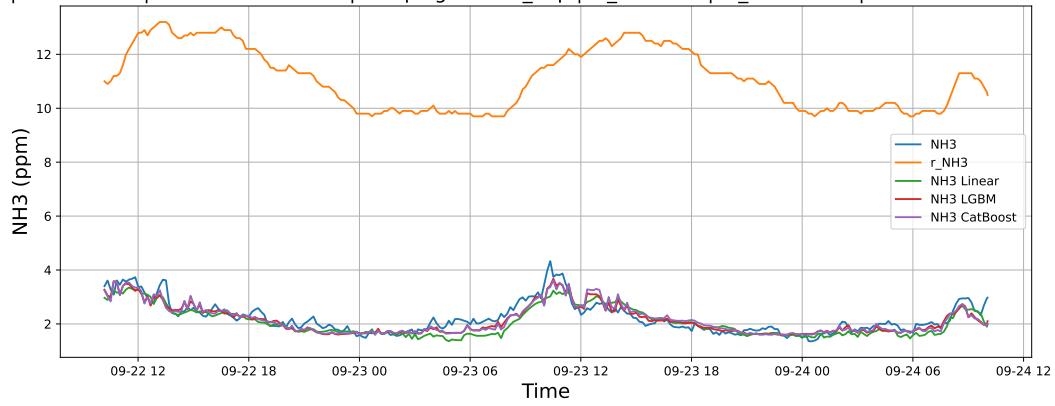
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



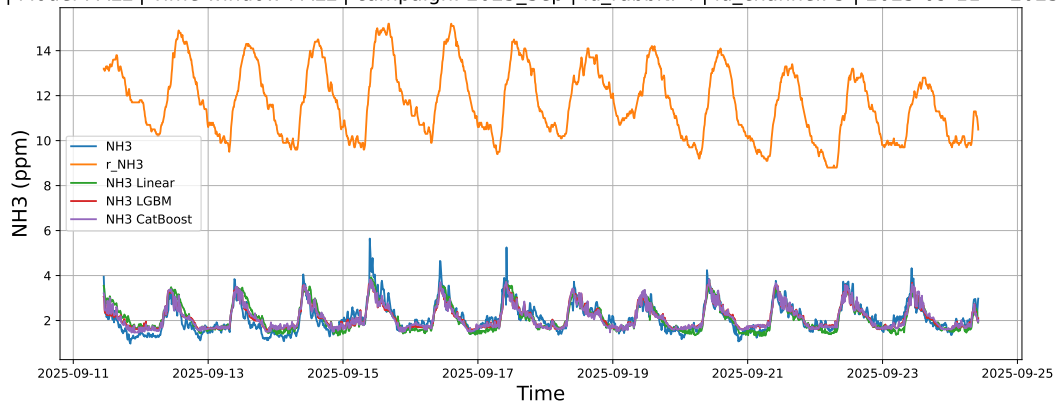
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

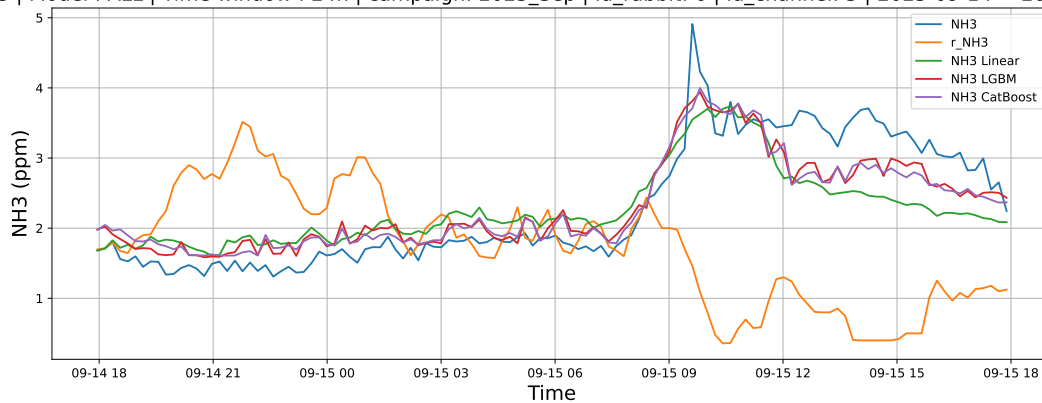
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

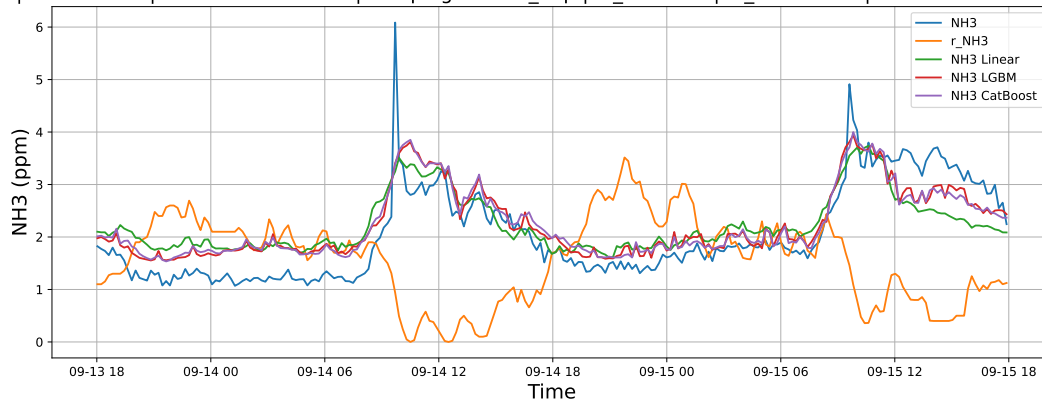
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

